

18 Striped red mullet the North Sea, eastern English Channel, Skagerrak, and Kattegat

mur.27.3a47d – Mullus surmuletus in Subarea 4 and divisions 7.d and 3.a

This striped red mullet stock is under a biennial advice. No TAC is set for this stock.

In 2021, advice on fishing opportunities was again requested following the precautionary approach. Due to incomplete survey sampling in 2020 (there were issues with calculation of survey indices), a lack of length and age samples from the main fleets (including other areas and nations), and problems with model formulation, the ICES stock category of striped red mullet in Subarea 4 and divisions 7.d and 3.a was downgraded from category 3 to category 5. ICES advice on fishing opportunities for 2022 and 2023 was based on the average ICES catches (considering discards negligible) over the period 2004–2020. Based on length-based indicators (LBI) analysis, fishing mortality is estimated above MSY reference points, the stock size relative to reference points is unknown. For those reasons, the precautionary buffer was applied.

In 2023, the stock was benchmarked to try a SPiCT model (ICES, 2023b) and a new survey exploitable biomass index was developed combining survey data from Q3 and Q4. The benchmark concluded that SPiCT was not appropriate to assess the North Sea striped red mullet stock and recommended to apply the *chr* rule (ICES, 2023a) for category 3 stocks using the combined Q3–4 survey index and length frequency from commercial landings as input data. Since 2023, ICES advice on fishing opportunities are based on the *chr* rule.

In 2015, highest landings were observed, and since then landings have gradually decreased until 2018. In 2019, landings increased reaching nearly the level of 2015. This is mainly due to the exploitation of the strong 2018 cohort. Landings decreased thereafter, however, in 2023, the third highest amount of landings from the last ten years were observed. The structure of the population is still truncated and recent catches of this stock mainly consist of age 0 and age 1 fish. The fishery for striped red mullet would benefit from improved technical measures such as sorting grids, increased mesh size, and spatial and temporal closures. These measures could reduce the catches of small fish and contribute to more stable yields.

18.1 General

Ecosystem aspects and stock identity

Striped red mullet (*Mullus surmuletus*) is a benthic species. Young fish are distributed in coastal areas, while adults have a more offshore distribution. Benzinou *et al.* (2013) conducted stock identification studies based on otolith and fish shape in European waters and showed that striped red mullet can be geographically divided into two units: a Western Unit (subareas 6 and 8, and divisions 7.a–c, 7.e–k, and 9.a) and a Northern Unit (Subarea 4 (North Sea) and divisions 7.d (eastern English Channel) and 3.a (Skagerrak, Kattegat)).

A recent review of striped red mullet stock structure in the greater North Sea was realized by Cefas and presented to WGNSSK 2020 (Ellis, 2020). This review does not support the current stock definition used by ICES. Indeed, survey data from North Sea International Bottom Trawl Survey (NS-IBTS) might indicate that striped red mullet in Division 3.a should be considered as a separate stock. In addition, survey data and commercial data have highlighted migration patterns between the western English Channel and the southern North Sea, with striped red mullet

concentrating and mixing in the southern North Sea during summer. Thus, the assessment of striped red mullet in Subarea 4 and divisions 7.d–7.e may need to be assessed as a single stock or a stock complex with two subpopulations mixing during summer.

In the English Channel, the first sexual maturity was identified on fish of 16.2 cm for males and 16.9 cm for females (Mahé *et al.*, 2013). Juveniles are found in waters of low salinity, while adults are found at high salinity. Striped red mullet prefers sandy sediments (Carpentier *et al.*, 2009).

Adult striped red mullet feed on small crustaceans, annelid worms and molluscs, using their chin barbels to detect prey and search the mud.

18.2 Fisheries

Historically, France has taken most of the landings with a targeted fishery for striped red mullet (> 90% of landings at the beginning of the 2000s). This French fishery targeting striped red mullet is conducted by bottom trawlers using a mesh size of 70–99 mm in the eastern English Channel and in the southern North Sea.

The eastern English Channel and southern North Sea areas are also fished by trawlers of various types targeting a variety of species. Striped red mullet might be a bycatch in these fisheries.

From 2000, a Dutch targeted fishery with fly shooters, and a UK fisheries has also developed. Landings are shared by these three fleets in the most recent years.

18.3 ICES historical advice

Advice for 2024 and 2025

ICES framework for category 3 stocks was applied (*chr* rule, Method 2.2; ICES, 2022). The advice is based on the *chr* rule to provide MSY advice. A relative exploitable biomass index from the combined NS-IBTS Q3, BTS Q3, and FR-GFS Q4 surveys was used as an indicator of stock development. The advice is based on the stock indicator for 2022, multiplied by a constant harvest rate, a biomass safeguard, and a precautionary multiplier. The stability clause was not applied, since the change from the previous advice was between +20% or –30%. Discarding is considered negligible.

ICES advises that when the MSY approach is applied, catches should be no more than 1 985 tonnes in each of the years 2024 and 2025. All catches are assumed to be landed.

Advice for 2022 and 2023

The ICES framework for category 5 stocks was applied (ICES, 2023a). For stocks without information on abundance or exploitation, ICES considers that a precautionary reduction of catches should be implemented where there is no ancillary information clearly indicating that the current level of exploitation is appropriate to the stock. Discarding is considered negligible.

Fishing mortality is above proxies of the MSY reference points (as indicated by a length-based analysis). The stock size relative to reference points is unknown. For these reasons, the precautionary buffer, which was last applied in 2017, was applied again in this assessment.

ICES advises that when the precautionary approach is applied, catches should be no more than 1950 tonnes in each of the years 2022 and 2023. All catches are assumed to be landed.

Advice for 2020 and 2021

ICES has not been requested to provide advice on fishing opportunities for this stock.

Advice for 2018 and 2019

ICES advises that the fishery for striped red mullet should be managed through technical measures that would reduce the catches of small fish and would contribute to more stable yields.

Fishing mortality is above proxies of the MSY reference points (as indicated by a length-based analysis). The stock size relative to reference points is unknown. For these reasons, the precautionary buffer, which was last applied in 2013, was applied again in this assessment.

ICES advises that when the precautionary approach is applied, catches should be no more than 465 tonnes in each of the years 2018 and 2019. All catches are assumed to be landed.

18.4 Management

No specific management objectives are known to ICES. There is no TAC for this species.

There is no minimum landing size for this species.

Demersal fisheries in the area are mixed fisheries, with many stocks exploited together in various combinations in different fisheries. In these cases, management advice must consider both the state of individual stocks and their simultaneous exploitation in demersal fisheries. Stocks in the poorest condition, particularly those which suffer from reduced reproductive capacity, become the overriding concern for the management of mixed fisheries, where these stocks are exploited either as a targeted species or as a bycatch.

18.5 Data available

18.5.1 Catch

Official landings data are shown by country in Table 18.5.1.1, Figure 18.5.1.1 and by area in Table 18.5.1.2, Figure 18.5.1.2. There is no indication that striped red mullet is discarded. All catches are assumed to be landed. Before 2000, most of the landings are recorded by France, with the first peak in landings occurring in 1995 with a total of 2079 tonnes. In 1999, France did not report its landings and only 72 tonnes were reported in total that year. Data were corrected based on the French national database value and ended up to be 1255 tonnes in total for 1999. From 2010 to 2013, the stock declined and the catches dropped to 364 tonnes. Up until 2012, France remained the main fleet landing striped red mullet. Thereafter, The Netherlands became the main country fishing on this stock. At the same time, the UK landings also increased to become the second most important country landing striped red mullet since 2020. In 2024, The Netherlands landed 47% of the total landings, UK 29%, France 20%, the remaining 4% are mainly caught by Belgium. Most of the landings come from ICES Division 7.d. In the period 1975 to 2000, the contribution of Subarea 4 to the total landing has gradually increased. In 2024, 35% of the landings originated from Subarea 4 and 65% from Division 7.d.

Table 18.5.1.3 presents total official landings and ICES estimates over the period 2004–2024 as well as the predicted catch corresponding to advice. Figure 18.5.1.3 shows that IC data and official landings are consistent over the years, except in 2008 where official landings are two times higher than ICES landings. This discrepancy is explained by an issue with French officially reported landings in that given year. Figure 18.5.1.4 shows the proportion of total ICES landings by country and displays the same changes as observed in official landings, including the change in main countries targeting striped red mullet. Figure 18.5.1.5 presents the changes in main gear used to catch the species. The Netherlands uses demersal seines, while France was mainly using bottom trawls. Since 2008, the proportion of landings from seine gears has increased. In recent

years, France also started to use demersal seines to target striped red mullet. In 2024, 74% of the catches were made using demersal seines and 22% using demersal trawls.

In 2025, UK resubmitted 2021-2023 data to InterCatch and the Netherlands resubmitted 2022 data. As shown in Figure 18.5.1.6, this resulted in minor changes in ICES landings with a downward revision of 2021-2022 ICES landings and upward revision of 2023 ICES landings.

18.5.2 Catch-at-age

Total landings were provided under the ICES InterCatch format for the period 2004–2024. However, only France provided age compositions to ICES InterCatch and mostly from Division 7.d. Although age structured landings are no longer used in the assessment, they are still requested in the data call to provide some information on the cohort strength. Figure 18.5.2.1 shows that in 2024, 10% of landings in Division 7.d were provided with an age structure and only 1.6% of landings in area 4 were provided with an age structure. Overall 7% of 2024 landings were provided with age structure.

As discards are considered negligible, they are not raised in InterCatch. Age allocation in InterCatch was done quarterly when possible, if not by semester. As mentioned in section 18.5.1, data were re-uploaded for 2021-2023 and the same allocation schemes as previous years were applied. The table below summarizes the allocation scheme used for 2021-2024 ICES landings. The overall age distribution has not changed after the data resubmission, only the total number of landings per year has changed (Figure.18.5.2.2).

Year	Unsampled strata				Year
	Q1	Q2	Q3	Q4	
2021	Q1		Q4		All
2022	Q1		Q3	Q4	All
2023	Q1		Q4		All
2024	Q1		Q4		All

Prior to 2009, no landings of age 0 were observed (Figure 18.5.2.3, and Table 18.5.2.1). Most of the landings are age 1. There is no age reading problem reported. This change in the landings might reflect a change in the reporting or a change in the fishing behaviour.

Mean weights at age were computed as described in the Stock Annex and are presented in Figures 18.5.2.4 and Table 18.5.2.2. Weights at age in the landings show a slight decrease for the oldest ages. However, sampling intensity for these ages is very low due to the small number of fishes in the catches.

18.5.3 Catch-at-length

Length frequency distributions are available in InterCatch from 2014 onwards. Due to the lack of sampled fleets in 2017, data of this year were not used in the assessment. As for age structured landings, length samples are mostly provided by France. In addition, few samples are provided by the UK and The Netherlands. Figure 18.5.3.1 shows that in 2024, 26% of landings in Division 7.d were provided with length distributions compared to 15% in area 4. Overall, 22% of 2024 landings were provided with length distributions. Length frequency samples are allocated

quarterly when possible. As mentioned in section 18.5.1, data were re-uploaded for 2021-2023 and the same allocation schemes as previous years were applied. The table below summarizes the allocation scheme used for 2021-2024 ICES landings. The overall length distribution has changed slightly after the data resubmission, as well as the total number of landings per year (Figure.18.5.3.2).

Year	Unsampled strata				Year
	Q1	Q2	Q3	Q4	
2021	Q1	Q1&2		Q4	All
2022	Q1	Q2	Q3	Q4	All
2023	Q1	Q2	Q3	Q4	All
2024	Q1	Q2	Q3	Q4	All

In an attempt to increase the length of the time-series, the French national dataset was analysed in 2023. Length samples are available from 2004 to 2022. Before 2014, most of the years in the French dataset have enough sampled fleets to be used to allocate ICES landings, except for 2008 and 2013 which were removed from the dataset (Figure 18.5.3.3). Figure 18.5.3.4 presents the French sampling coverage compared to the French landings and the total ICES landings, and also the contribution of each ICES Subarea and division to the total ICES landings. Except for 2008, the sampling coverage is always above 20% in Division 7.d and remains small in area 4 as observed also in the InterCatch data. Based on the number of samples available per quarter for each year in the French dataset (Figure 18.5.3.3), the following allocation procedure was used:

Year	Unsampled strata			
	Q1	Q2	Q3	Q4
2004	Q2 & 3	Q2 & 3	Q3	Q2 & 3
2005	Q2 & 3	Q2 & 3	Q3	Q2 & 3
2006	Q3 & 4	Q3 & 4	Q3	Q4
2007	Q1,3 & 4	Q1,3 & 4	Q3	Q1,3 & 4
2009	Q1 & 2	Q1 & 2	Q3	Q4
2010	Q1 & 2	Q2	Q3	Q4
2011	Q2	Q2	Q3	Q3 & 4
2012	Q1 & 2	Q2	Q3	Q4

The resulting length distribution is shown in Figure 18.5.3.5.

18.5.4 Survey data

18.5.4.1 Survey index defined during the last benchmark

During the last benchmark in 2023 (ICES, 2023b), the NS-IBTS-Q3 (International Bottom Trawl Survey), BTS-Q3 (Beam Trawl Survey) and FR-GFS Q4 (French channel Ground Fish Survey)

surveys were estimated to be good indicators of the population trends as they cover the spatial distribution of this stock. A detailed description of model selection can be found in WKBM-SYSPIC2 benchmark report (ICES, 2023b). In 2015, there was a change in the research vessel used for the FR-GFS. The consequences of these changes were assessed via an intercalibration in 2014 and an analysis of the catch data (ICES, 2017, Section “CGFS: Change of vessel from 2015 onwards and consequences on survey design and stock indices”). This revealed that for striped red mullet, indices could be used without a correction factor. During the benchmark, an exploitable biomass index was developed by correcting the survey selectivity against commercial selectivity. All individuals below 13 cm were removed from the dataset as only 2.5% of the commercial catches (2014–2021) were below 13 cm. To accommodate for areas with no occurrence of striped red mullet, the northern part of the North Sea was removed from the survey dataset as shown in Figure 18.5.4.1 and 18.5.4.2. Most of the abundance is located in the southern North Sea and in the eastern English Channel. The BTS catches smaller individuals than the NS-IBTS and only few individuals larger than 12 cm. Figure 18.5.4.3 illustrates that a large amount of striped red mullet is located in Division 7.d. As the FR-GFS is using the same gear as the NS-IBTS, the difference in abundance can be explained either by the difference in timing between the two surveys (Q3vs.Q4) or the difference in location.

The standardized exploitable biomass index was developed to combine the three surveys using a tweedie GAM model, accounting for changing gear, season and area. The following model was adjusted using the SurveyIndex package in R (Berg *et al.*, 2014):

$$g(\mu_i) = \text{Year}_i + \text{Gear}_i + U(\text{Country}/\text{Ship}_i) + f_1(\text{lon}_i, \text{lat}_i) + f_2(\text{depth}_i) + f_3(\text{Time of Year}_i) + f_4(\text{Time of day}_i) + f_5(\text{lon}_i, \text{lat}_i, \text{Year}_i) + \text{offset}(\log(\text{Swept_area}_i))$$

where for each haul i ,

μ_i is the expected response (biomass for the Tweedie distribution)

Year_i a categorical year effect

Gear_i a categorical gear effect

U is a random Country-Ship effect

f_1 is a 2-D thin plate regression spline (“ts”) on the latitude/longitude coordinates

f_2 is a 1-D thin plate spline (“ts”) for the bottom depth effect

f_3 and f_4 are cubic regression spline (“cc”) for the Time of Year and Time of day

f_5 is a tensor product interaction between a two-dimensional thin plate regression spline on the latitude/longitude coordinates and a cubic regression spline (“cs”) on the year

g is the link function (log transformed response)

$\log(\text{Swept_area}_i)$ is an offset to standardize the response by the swept-area.

18.5.4.2 Survey index estimates and diagnostics

Figure 18.5.4.4 presents survey index model residuals against all covariates. Most of the different effects are captured by the model and no obvious trend is observed in the residuals even if few strong residuals (above 4) can be observed. They are mostly observed in shallow water (below 50 m) of the UK and France before 2014. Figure 18.5.4.5 and Table 18.5.4.1 present the standardized survey index. The index is consistent with what is observed in the catches. From 2009–2013, the stock was at a low level as seen in the decline of the catches in Figure 18.5.1.1. Since this period of low exploitable biomass, three high peaks were estimated in 2015, 2019 and 2023. In 2024, the index decreased just below the level of 2022. The leave-one-out analysis in Figure 18.5.4.6 shows a consistent trend when removing one survey at a time. The main noticeable

changes occur when removing the FR-GFS survey, with the 2014-2015 peak no longer appearing and a drop in 1997 and 2018. In any case, the last 2019 and 2023 peaks of exploitable biomass are observed and the trend over the last 6 years remains within the confidence interval. As displayed in Figure 18.5.4.7, no retrospective bias is observed in the survey index, especially in recent years. The Mohn's Rho calculated on the last 5 peels is 0.129.

18.6 Assessment

18.6.1 Category 3 advice 2026–2027: *chr* rule (Method 2.2)

Since the last benchmark (ICES, 2023b), striped red mullet assessment in Subarea 4 and divisions 7.d and 3.a is based on the category 3 *chr* rule (Method 2.2). The assessment uses length data available from 2004 onwards, with data originating only from the French national database before 2014 and provided in InterCatch thereafter. An exploitable biomass index combining NS-IBTS Q3, BTS Q3 and FR-GFS Q4 was available from 1991–2024. During the benchmark (ICES, 2023b), growth parameters were estimated using all age sampling data available in the French database (commercial data and survey data from 2006–2021). The k was estimated to be equal to 0.36 for females and 0.42 for males and L_{∞} at 36.6 cm and 27.8 cm for females and males respectively. When combining both sexes, k is estimated at 0.36 and L_{∞} at 34.1 cm.

When k is found to be equal or larger than 0.32 and smaller than 0.45, the *chr* rule needs to be applied with $m = 0.5$.

The *chr* formula contains different factors to determine the catch in the advice year:

$$C_{y+1} = I_{y-1} \times F_{proxyMSY} \times b \times m$$

where

- $y = 2025$
- C_{y+1} = the catch in the advice year, 2026
- I_{y-1} = the index value of 2024. For striped red mullet we used a standardized exploitable biomass index estimated from NS-IBTS Q3, BTS Q3 and FR-GFS Q4 shown in Table 18.5.4.1.
- m = the multiplier applied to the harvest control rule to maintain the probability of the biomass declining below B_{lim} to less than 5%. This multiplier is set to 0.5 for the *chr* rule.
- b = the biomass safeguard. The most recent index data I_{2024} needs to be compared to $I_{trigger}$ with $I_{trigger} = 1.4 \times I_{loss}$ and I_{loss} = the lowest observed index value. As the stock size indicator of striped red mullet is estimated from a statistical model, I_{loss} and $I_{trigger}$ need to be estimated every time advice is given for this stock to account for potential retrospective bias. For striped red mullet $I_{loss} = 20.03$, being the index value in 2013 (Table 18.5.4.1); however during the period 2009 to 2013 the stock size and the catch has declined and in 2013 the stock had likely collapsed. Consequently, it was decided to set I_{loss} at the 20th percentile of the index estimated values between 1991 and 2022, giving $I_{loss} = 76.62$. $I_{trigger}$ is therefore = 107.27, which is smaller than $I_{2024} = 266.48$ (Figure 18.6.1.1). Therefore, b should be set equal to 1.
- $HR_{proxyMSY}$ = the harvest rate for the set of years for which the fishing pressure proxy relative to the MSY proxy indicator ratio $L_{mean} / L_F = M$ is larger than 1. We calculated the $HR_{proxyMSY}$ in different steps. The same as for $I_{trigger}$, to account for potential retrospective bias in the stock size indicator, $HR_{proxyMSY}$ must be revised each time advice is given for this stock.
 - First, we calculated the length at first capture (Lc) by year, which is defined as the first length class where abundance is more than or equal to half of the maximum

abundance. Length data are available through InterCatch from 2014 onwards and from the French national database from 2004 to 2013. Only landings are used and years with not enough samples were dropped from the calculation as shown in the table below. L_c was defined as the average of yearly L_c over the period 2004-2022 at 178.75 mm (Figure 18.6.1.2). Even with the resubmission of the 2021-2022 landings data in 2025, no change in L_c was observed.

Year	L_c (mm)
2004	190
2005	180
2006	150
2007	190
2008	-
2009	190
2010	150
2011	190
2012	200
2013	-
2014	130
2015	190
2016	190
2017	-
2018	200
2019	180
2020	180
2021	190
2022	160
2023	160
2024	160
mean (2004-2022)	178.75

- Second, L_∞ (length infinity) is estimated at 341 mm for both sexes as mentioned above.
- Third, the target reference length $L_{F=M} = 0.75L_c + 0.25L_{inf} = 219.31$ mm
- Fourth, the mean length above L_c is calculated (Figure 18.6.1.3). Minor changes in L_{mean} are observed from 2021-2023 due to InterCatch landings data resubmission.

Year	Lmean > Lc (mm)
2004	207.85
2005	233.68
2006	229.39
2007	211.81
2008	-
2009	225.54
2010	209.56
2011	211.28
2012	230.63
2013	-
2014	213.82
2015	210.96
2016	219.27
2017	-
2018	225.24
2019	204.32
2020	204.06
2021	209.22
2022	204.54
2023	204.72
2024	211.90

- Fifth, the quantity inverse f was calculated as the ratio of $L_{F=M}$ and the mean length above L_c . Up to 2022, only for the years 2005, 2006, 2009, 2012 and 2018, the $F_{MSY \text{ proxy}}$ indicator relative to the fishing pressure proxy ratio $L_{F=M} / L_{mean}$ (inverse f) was lower than 1 (Figure 18.6.1.4).
- Sixth, the harvest rate (ratio of catch over index) was calculated (Figure 18.6.1.4).
- Finally, the HR_{proxyMSY} being the mean of the harvest rates for the years 2005, 2006, 2009, 2012 and 2018 (where up to 2022 quantity inverse $f < 1$) was updated to 14.50 (Figure 18.6.1.4).
- Below is a table summary of all the indicators, they were all computed and plotted using the cat3advice R package from GitHub¹.

¹ <https://github.com/shfischer/cat3advice>

Year	Lc (mm)	Lc av. (mm)	Linf (mm)	L _{F=M} (mm)	L _{mean} > Lc (mm)	Quantity inverse <i>f</i> (L _{F=M} /L _{mean})	Index (I)	Catch (C, tons)	Harvest rate (C/I)
2004	190	178.75	341	219.3125	207.8474847	1.0551607	267.143766	4668.8481	17.4769120
2005	180	178.75	341	219.3125	233.6755248	0.9385343	132.078396	2347.1633	17.7709858
2006	150	178.75	341	219.3125	229.3942461	0.9560506	170.835866	1475.764	8.63849045
2007	190	178.75	341	219.3125	211.8140799	1.0354010	563.577661	4604.1779	8.16955359
2008	-	-	-	-	-	-	69.7830859	2063.8964	29.5758832
2009	190	178.75	341	219.3125	225.5402677	0.9723873	113.293720	1513.0439	13.3550553
2010	150	178.75	341	219.3125	209.5627129	1.0465244	74.4794484	1919.3781	25.7705735
2011	190	178.75	341	219.3125	211.2784324	1.0380260	44.4950169	1511.4629	33.9692623
2012	200	178.75	341	219.3125	230.6336748	0.9509127	32.8775635	725.7319	22.0737738
2013	-	-	-	-	-	-	20.2976812	407.8162	20.0917630
2014	130	178.75	341	219.3125	213.8237972	1.0256693	360.546654	1717.63681	4.76397935
2015	190	178.75	341	219.3125	210.9550875	1.0396170	408.450713	4486.5694	10.9843593
2016	190	178.75	341	219.3125	219.2710576	1.0001890	103.798757	2578.72175	24.8434744
2017	-	-	-	-	-	-	140.575798	2195.23362	15.6160139
2018	200	178.75	341	219.3125	225.2414619	0.9736773	153.916538	1639.75534	10.6535357
2019	180	178.75	341	219.3125	204.3175364	1.0733905	417.688586	4047.50848	9.69025398
2020	180	178.75	341	219.3125	204.0638604	1.0747248	293.353745	3503.04775	11.9413773
2021	190	178.75	341	219.3125	209.2226847	1.0482252	219.555208	2612.3349	11.8983055
2022	160	178.75	341	219.3125	204.5454668	1.0721944	280.460862	2893.4238	10.3166759
2023	160	178.75	341	219.3125	207.7199003	1.0558088	447.925673	3941.8213	8.80016825
2024	160	178.75	341	219.3125	211.8981656	1.0349901	266.480176	3289.1247	12.3428496

The $f=L_{\text{mean}}/L_{F=M}$ remains the basis for selecting a HR_{proxyMSY} (the mean of historical C/I values for which $f>1$ is selected or inverse $f < 1$), and it therefore remains the principal motivation for acceptable levels of fishing exploitation in the past. It therefore makes sense to use inverse f as a basis for showing fishing pressure (exploitation status), which would make this consistent across category 3 stocks (based on *rfb* or *chr* rules). HR_{proxyMSY} is really a means to get from current stock index value to catch advice that is consistent with acceptable observed catch levels in the past, and an HR plot should therefore not be shown as indicating fishing pressure in the advice (although it could be shown in the report) - showing both an HR plot and an inverse f plot would cause unnecessary confusion in the advice, and they would often be conflicting. The terminology of naming the catch-index ratio a “harvest rate” should be reconsidered by WKLIFE.

18.6.2 2025 advice

Advice for 2026 and 2027

The ICES framework for category 3 stocks was applied (ICES 2023a). Fishing pressure on the stock is above $F_{MSY\ proxy}$ (Figure 18.6.1.4) and the stock size indicator is above $I_{trigger}$ (Figure 18.6.1.1).

ICES advises that when the MSY approach is applied, catches in each of the years 2026 and 2027 should be no more than 1932 tonnes. All catches are assumed to be landed.

Previous catch advice A_y (advised catch for 2024 and 2025)	1985 tonnes
Biomass index	
I : most recent biomass index (I_{2024})	266.48
MSY proxy harvest rate	
$HR_{MSY\ proxy}$: MSY proxy harvest rate (average of the ratio of catch to biomass index for the years for which $f > 1$, where $f = L_{mean}/L_{F=M}$)	14.50
Biomass safeguard	
Index trigger value ($I_{trigger}$)	107.27
b : index relative to trigger value, $\min\{I_{2024}/I_{trigger}, 1\}$	1
Precautionary multiplier to maintain biomass above B_{lim} with 95% probability	
m : multiplier (generic multiplier based on life history)	0.5
CHR calculation**	1932 tonnes
Stability clause (+20%/–30% compared to A_y , only considered if $b=1$)	Not applied
Discard rate	0%
Catch advice A_{y+1} for 2026 and 2027**	1932 tonnes
Landings corresponding to the advice for 2026 and 2027 ***	1932 tonnes
% advice change	–2.7%

* The figures in the table are rounded. Calculations were done with unrounded inputs, and computed values may not match exactly when calculated using the rounded figures in the table.

** $A_{y+1} = I \times HR_{MSY\ proxy} \times b \times m$, limited by stability clause if applicable.

*** $A_{y+1} \times (1 - \text{Discard rate})$

18.7 Length-based indicators screening

The LBI's were computed using the length distributions shown in Figure 18.5.3.5. Average L_c and L_{mean} and $L_{F=M}$ were estimated the same way as for the *chr* rule (section 18.6.1, Figures 18.6.1.2 and 18.6.1.3).

Most of the indicators appear outside the established references in 2024 (see table 18.7.1):

- Yearly length at first catch L_c and length of 25% of catches are above L_{mat} (16.9 cm) most of the time. However in recent years, L_c is below L_{mat} .
- Ratio of the L_{max5} , mean length of the 5% largest catches, to L_{inf} (40 cm) is around 0.7/0.8 most of the time, which clearly shows the lack of big/old fish in the population.
- Using an average L_c , L_{mean}/L_{opt} is around 0.95 and shows an exploitation close to optimal yield even if only few large fish remain in the stock.
- $L_{mean}/L_{F=M}$ below 1 tends to show that this stock is not exploited sustainably in most recent years, with the exception of 2018, where the ratio is just above 1.

This indicates that the stock is considered not to be exploited sustainably. The main concerns are for the big/old fish that are missing from the population. Length-based indicators based on samples from commercial catches show that in relation to conservation criteria there is evidence of growth overfishing, meaning the fish is caught before it has realized its growth potential (Table 18.7.1).

18.8 Conclusions

Good recruitment was observed in the catches in 2015, 2019 and in 2023 based on landings at age and length frequency distributions. There is no TAC for this species, so the advice was not implemented and the catches overshot the advice from 2014 onwards. In 2024, the advice was over-shot by 1304 tonnes corresponding to +66% compared to the advice for 2024. The stock age distribution appears to be still truncated and landings are mainly composed of age 1.

Length-based indicators based on samples from commercial catches (2004–2024) showed in 2024 that in relation to conservation criteria there is evidence of growth overfishing, meaning the fish are caught before they have realized their growth potential. The exploitable biomass depends on recruitment.

18.9 Issues list

18.9.1 Data and stock ID

- Age data from other countries than France need to be provided as everything is actually raised using the French catches in the eastern Channel and few samples from the North Sea.
- Length data were mainly provided by France and partly by the UK, with recently few submissions from The Netherlands. However the sampling coverage remains low and higher coverage is required.
- Age or Length based survey indices should be explored in future to allow for the use of a stage-biomass model or an age based model.
- Even if discards are expected to be very low (no minimum landing size, high price), discard data should be re-investigated.
- Based on Ellis (2020) stock ID should be reinvestigated.

18.9.2 Assessment

- Explore methods applied to "short lived species" (two stages model)?
- Other models should also be explored (SS3)

18.9.3 Forecast and reference points

- This stock is not category 1, so no forecast is currently done. This should be investigated if the assessment method is improved. However, there is no TAC for the stock so a forecast is not a priority, although reference points are still important.

18.10 References

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18.11 Figures and tables

Table 18.5.1.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Official landings by country (tonnes).

Year	Belgium	Denmark	France	Germany	Netherlands	UK	Total
1975	0	0	140	0	0	0	140
1976	0	0	156	0	3	1	160
1977	0	0	279	0	12	1	292
1978	0	0	207	0	25	3	235
1979	0	0	212	0	32	11	255
1980	0	0	86	0	25	4	115
1981	0	0	44	0	19	1	64
1982	0	2	32	0	18	2	54
1983	0	0	232	0	15	1	248
1984	0	0	204	0	0	3	207
1985	0	1	135	0	0	4	140
1986	0	1	84	0	0	3	88
1987	0	2	40	0	0	3	45
1988	0	1	35	0	0	4	40
1989	0	0	37	0	0	5	42
1990	0	0	524	0	0	13	537
1991	0	0	208	0	0	11	219
1992	0	0	431	0	0	14	445
1993	0	0	516	0	0	18	534
1994	0	0	308	0	0	14	322
1995	0	0	2016	0	0	63	2079
1996	0	1	1785	0	1	36	1823
1997	0	1	731	0	0	48	780
1998	0	1	2598	0	0	97	2696
1999 ¹⁾	0	2	1183	0	0	70	1255
2000	0	2	2590	13	235	93	2933
2001	0	5	1417	10	533	142	2107

Year	Belgium	Denmark	France	Germany	Netherlands	UK	Total
2002	0	12	1346	9	326	82	1775
2003	17	0	2750	15	396	115	3293
2004	22	0	3618	26	804	91	4561
2005	19	0	1595	26	600	81	2321
2006	12	0	1030	0	293	69	1404
2007	14	0	3475	0	906	161	4556
2008	16	0	3249	0	873	313	4451
2009	13	0	736	0	562	260	1571
2010	62	0	879	0	567	310	1818
2011	83	0	649	0	540	238	1510
2012	38	0	155	0	367	138	698
2013	33	0	112	0	180	39	364
2014	71	0	726	0	700	242	1739
2015	211	0	1615	0	2038	663	4527
2016	151	0	556	0	1421	486	2614
2017	93	0	784	0	978	349	2204
2018	77	0	593	0	826	156	1652
2019	232	0	1408	0	1867	589	4096
2020	220	0	723	0	1752	835	3530
2021	53	0	595	0	1188	757	2593
2022	79	4	790	0	1132	838	2843
2023	123	3	920	0	1747	1088	3881
2024	137	1	671	0	1537	940	3286

1) No data reported by France in 1999. The value reported in that table is extracted directly from French national database.

Table 18.5.1.2. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Official landings by area (tonnes). Note: Most of the Subarea 4 catches are realized in Division 4.c.

Year	3.a	4	7.d	Total ²⁾
1975	0	0	140	140
1976	0	4	156	160
1977	0	19	273	292
1978	0	30	205	235
1979	0	49	206	255
1980	0	29	86	115
1981	0	20	44	64
1982	0	21	33	54
1983	0	41	207	248
1984	0	22	185	207
1985	0	10	130	140
1986	0	6	82	88
1987	0	7	38	45
1988	0	7	33	40
1989	0	5	37	42
1990	0	33	504	537
1991	0	26	193	219
1992	0	30	415	445
1993	0	63	471	534
1994	0	58	264	322
1995	0	527	1552	2079
1996	0	264	1559	1823
1997	0	139	641	780
1998	0	389	2307	2696
1999 ¹⁾	0	251	1003	1254
2000	0	895	2038	2933
2001	0	810	1297	2107
2002	0	626	1149	1775

Year	3.a	4	7.d	Total ²⁾
2003	0	824	2469	3293
2004	0	936	3625	4561
2005	0	728	1593	2321
2006	0	321	1082	1403
2007	0	773	3782	4555
2008	0	915	3536	4451
2009	0	455	1115	1570
2010	0	350	1468	1818
2011	0	305	1206	1511
2012	0	193	505	698
2013	0	98	266	364
2014	0	263	1476	1739
2015	0	800	3727	4527
2016	0	824	1789	2613
2017	0	652	1552	2204
2018	0	385	1267	1652
2019	0	1308	2788	4096
2020	0	1387	2143	3530
2021	0	901	1692	2593
2022	0	1046	1797	2843
2023	0	1667	2215	3882
2024	0	1159	2125	3284

¹⁾ No data reported by France in 1999. The value reported in that table is extracted directly from French national database.

²⁾ Differs from Table 18.5.1.1 and Table 18.5.1.3 due to rounding.

Table 18.5.1.3 Striped red mullet in Subarea 4 and divisions 7.d and 3.a. History of ICES advice, the agreed TAC, and ICES estimates of landings. Weights in tonnes.

Year	ICES Advice	Predicted catch corresp. to advice	Official landings ¹⁾	ICES Estimates
2004			4561	4669
2005			2321	2347
2006		-	1404	1476
2007		-	4555	4604
2008		-	4451	2064
2009		-	1570	1513
2010		-	1818	1919
2011		-	1510	1511
2012	No increase in catch	-	698	726
2013	No increase in catches (average 2009–2010)	< 1700	364	408
2014	Reduce catches by 36% compared to 2012	< 460	1739	1718
2015	No new advice, same as for 2014	< 460	4527	4487
2016	Precautionary approach	<552	2613	2579
2017	Precautionary approach	<552	2204	2195
2018	Precautionary approach	<465	1652	1640
2019	Precautionary approach	<465	4096	4048
2020	No Advice	-	3531	3503
2021	No Advice	-	2593	2612
2022	Precautionary approach	<1950	2843	2893
2023	Precautionary approach	<1950	3882	3942
2024	MSY approach	<1985	3285	3289
2025	MSY approach	<1985		

¹⁾ Differs from Table 18.5.1.1 and Table 18.5.1.2 due to rounding.

Table 18.5.2.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Landing numbers-at-age (thousands).

	0	1	2	3	4	5	6	4+
2004	0	43076	1826	940	75	111	0	186
2005	0	16557	2448	262	56	199	0	255
2006	0	3900	2325	1674	109	78	0	187
2007	0	36872	1120	551	94	33	0	127
2008	0	1316	10459	1248	313	221	0	534
2009	45	13256	1075	540	83	0	0	83
2010	12971	13384	593	125	70	19	1	90
2011	0	9310	1453	639	76	4	0	80
2012	6	1337	1246	1479	181	2	0	183
2013	1170	2342	395	244	0	0	0	0
2014	9904	10556	1300	14	14	14	0	28
2015	1728	35360	5952	18	2	32	0	34
2016	38	3498	9680	2129	148	51	0	199
2017	872	10314	2974	1105	223	130	100	453
2018	511	6630	3017	234	140	0	0	140
2019	1582	31105	1511	466	119	0	0	119
2020	590	27386	512	31	0	0	0	0
2021	740	15714	2966	345	8	0	0	8
2022	3084	17912	4335	232	42	0	0	42
2023	5151	19252	5233	1065	7	11	0	18
2024	1535	15706	6171	357	0	0	0	0

Table 18.5.2.2. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Stock weights (kg).

	0	1	2	3	4	5	6	4+
2004	0	0.09	0.222	0.27	0.434	0.66	0	0.569
2005	0	0.105	0.172	0.3	0.383	0.419	0	0.411
2006	0	0.146	0.188	0.241	0.379	0.35	0	0.367
2007	0	0.107	0.313	0.422	0.446	0.677	0	0.506
2008	0	0.096	0.139	0.226	0.326	0.41	0	0.361

	0	1	2	3	4	5	6	4+
2009	0.046	0.07	0.16	0.177	0.423	0	0	0.423
2010	0.042	0.077	0.112	0.24	0.225	0.149	0.215	0.209
2011	0	0.052	0.15	0	0	0.323	0	0.016
2012	0.023	0.091	0.169	0.255	0.229	0.772	0	0.235
2013	0.025	0.063	0.118	0.115	0	0	0	0
2014	0.029	0.093	0.144	0.259	0.294	0.323	0	0.309
2015	0.038	0.1	0.114	0.37	0.42	0.187	0	0.2
2016	0.038	0.114	0.138	0.319	0.42	0.187	0	0.360
2017	0.038	0.114	0.138	0.319	0.42	0.187	0	0.260
2018	0.046	0.143	0.166	0.273	0.315	0	0	0.315
2019	0.033	0.111	0.144	0.158	0.156	0	0	0.156
2020	0.038	0.114	0.110	0.320	0	0	0	0
2021	0.040	0.112	0.146	0.234	0.400	0	0	0.400
2022	0.059	0.094	0.130	0.228	0.211	0	0	0.211
2023	0.049	0.117	0.167	0.317	0.312	0.336	0	0.327
2024	0.052	0.111	0.145	0.302	0	0	0	0

Table 18.5.4.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Q3-Q4 standardized biomass exploitable index from 1991 to 2024, used for advice for 2026–2027.

Year	Q3-Q4 Survey index estimates			
	Low	Value	High	CV
1991	29.15	47.15	91.76	0.287
1992	68.64	97.62	171.54	0.229
1993	59.02	85.2	159.31	0.248
1994	23.87	36.81	65.77	0.253
1995	149.74	208.38	329.76	0.197
1996	100.19	139.87	220.09	0.197
1997	91.09	126.09	195.2	0.191
1998	214.74	300.11	460.82	0.191
1999	74.39	108.33	171.11	0.208
2000	155.28	213.81	320.16	0.181

Year	Q3-Q4 Survey index estimates			
	Low	Value	High	CV
2001	66.85	92.09	143.03	0.19
2002	97.98	139.37	214.73	0.196
2003	180.83	244.06	356.28	0.17
2004	202.59	267.14	403.58	0.172
2005	97.4	132.08	202.73	0.183
2006	124.82	170.84	258.91	0.182
2007	439.04	563.58	820.78	0.156
2008	49.62	69.78	110.4	0.2
2009	79.56	113.29	174.83	0.197
2010	51.61	74.48	122.3	0.216
2011	30.71	44.5	71.85	0.212
2012	21.54	32.88	54.28	0.231
2013	12.96	20.3	36.11	0.256
2014	268.25	360.55	557.33	0.183
2015	294.9	408.45	617.68	0.185
2016	73.34	103.8	166.7	0.205
2017	99.48	140.58	227.37	0.207
2018	113.6	153.92	252.62	0.2
2019	303.04	417.69	650.13	0.191
2020	214.62	293.35	460.4	0.191
2021	157.38	219.56	356.92	0.205
2022	214.76	280.46	451.55	0.186
2023	347.09	447.93	689.81	0.172
2024	215.64	266.48	432.34	0.174

Table 18.7.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Traffic light table for length-based indicators. Conservation criteria for small fish: L_c (yearly length at first catch) and 25% percentile relative to L_{mat} (length at 50% maturity); and for large fish: mean length of the largest 5% in the catch ($L_{max5\%}$) relative to asymptotic length L_{inf} and the proportion of mega spawners (P_{mega}). Optimizing yield criterion: the mean length L_{mean} is compared to the theoretical length of optimal biomass (L_{opt}). MSY criterion: L_{mean} is compared to $L_{F=M}$, the MSY proxy. "Ref" indicates the reference criterion: green colour for meeting the criterion, and red flagging issues (e.g. dome-shaped vs. overexploitation). "Ref" indicates the criterion required for a green light. Each year is evaluated separately.

LBI update	Conservation				Optimizing Yield	MSY
	L_c/L_{mat}	$L_{25\%}/L_{mat}$	$L_{max5\%}/L_{inf}$	P_{mega}	L_{mean}/L_{opt}	$L_{mean}/L_{F=M}$
Ref	>1	>1	>0.8	>30%	~1(>0.9)	≥1
2004	1.12	1.07	0.83	0.04	0.91	0.95
2005	1.07	1.18	0.88	0.25	1.03	1.07
2006	0.89	0.89	0.79	0.05	1.01	1.05
2007	1.12	1.18	0.77	0.03	0.93	0.97
2009	1.12	1.24	0.73	0.01	0.99	1.03
2010	0.89	1.18	0.9	0.19	0.92	0.96
2011	1.12	1.01	0.8	0.03	0.93	0.96
2012	1.18	1.12	0.8	0.04	1.01	1.05
2014	0.77	1.24	0.89	0.17	0.94	0.97
2015	1.12	1.12	0.89	0.06	0.93	0.96
2016	1.12	0.89	0.77	0.02	0.96	1
2018	1.18	1.12	0.76	0.02	0.99	1.03
2019	1.07	1.18	0.8	0.06	0.9	0.93
2020	1.07	1.12	0.85	0.11	0.9	0.93
2021	1.12	1.07	0.75	0.02	0.92	0.95
2022	0.95	1.12	0.73	0.01	0.9	0.93
2023	0.95	1.12	0.79	0.04	0.91	0.95
2024	0.95	1.01	0.75	0.02	0.93	0.97

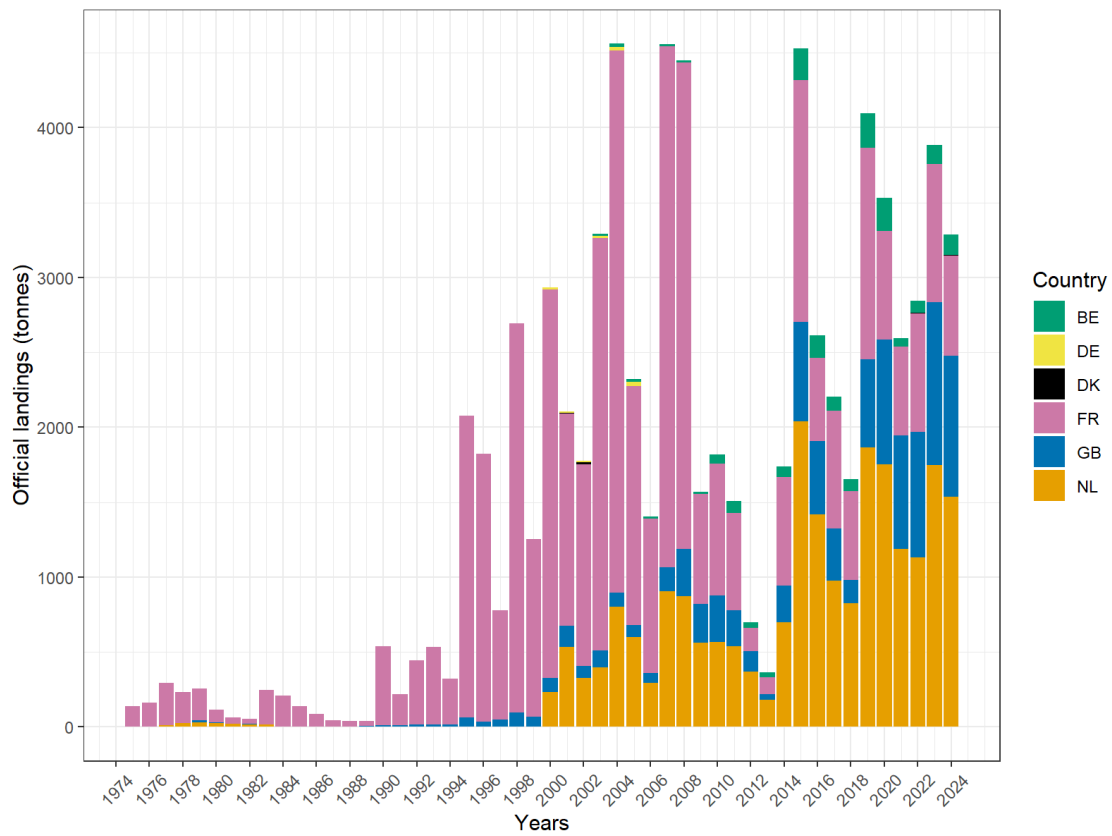


Figure 18.5.1.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Official landings by country (in tonnes). In 1999, French landings were corrected based on the national database value.

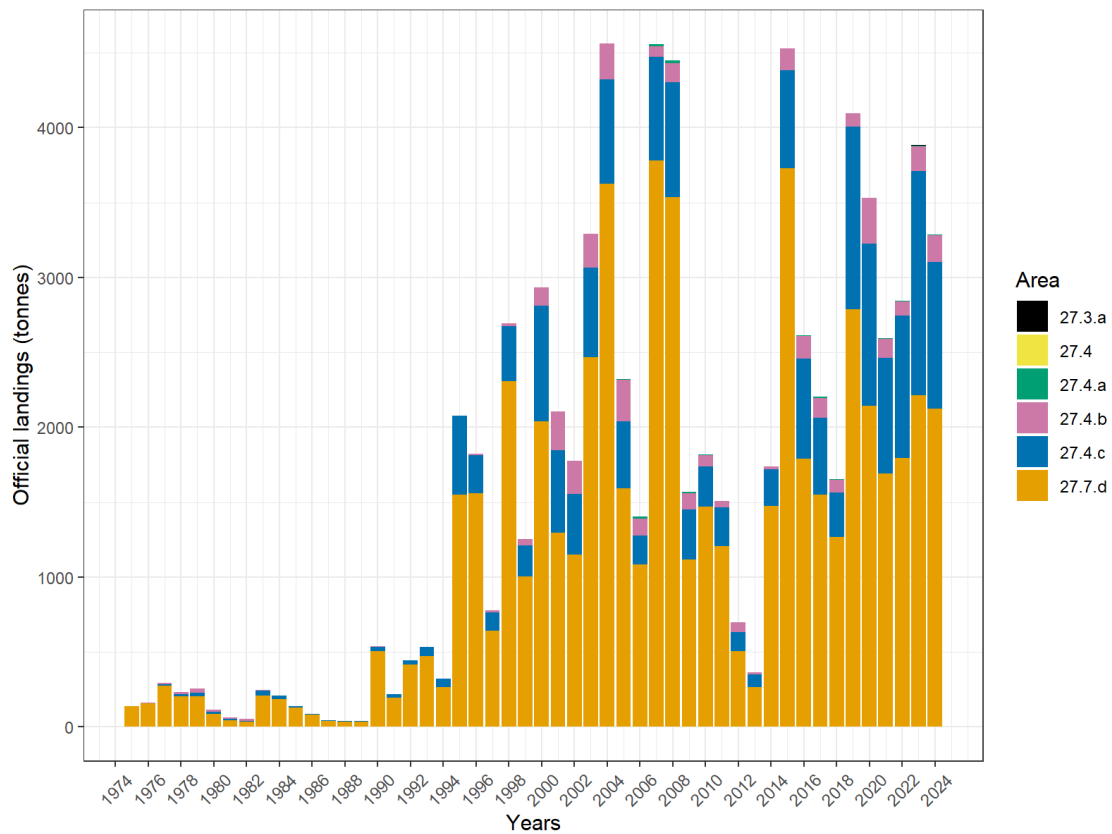


Figure 18.5.1.2. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Official landings by area (in tonnes). In 1999, French landings were corrected based on the national database value.

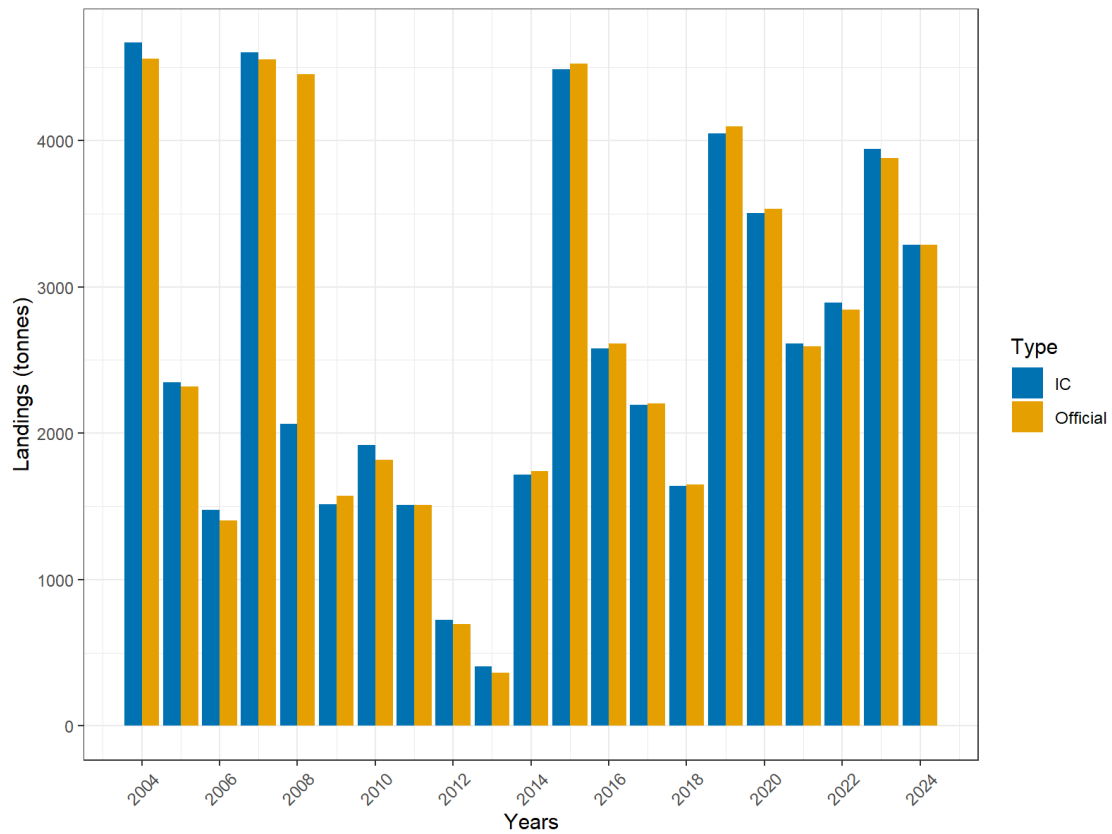


Figure 18.5.1.3. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Official landings compared to ICES landings (in tonnes).

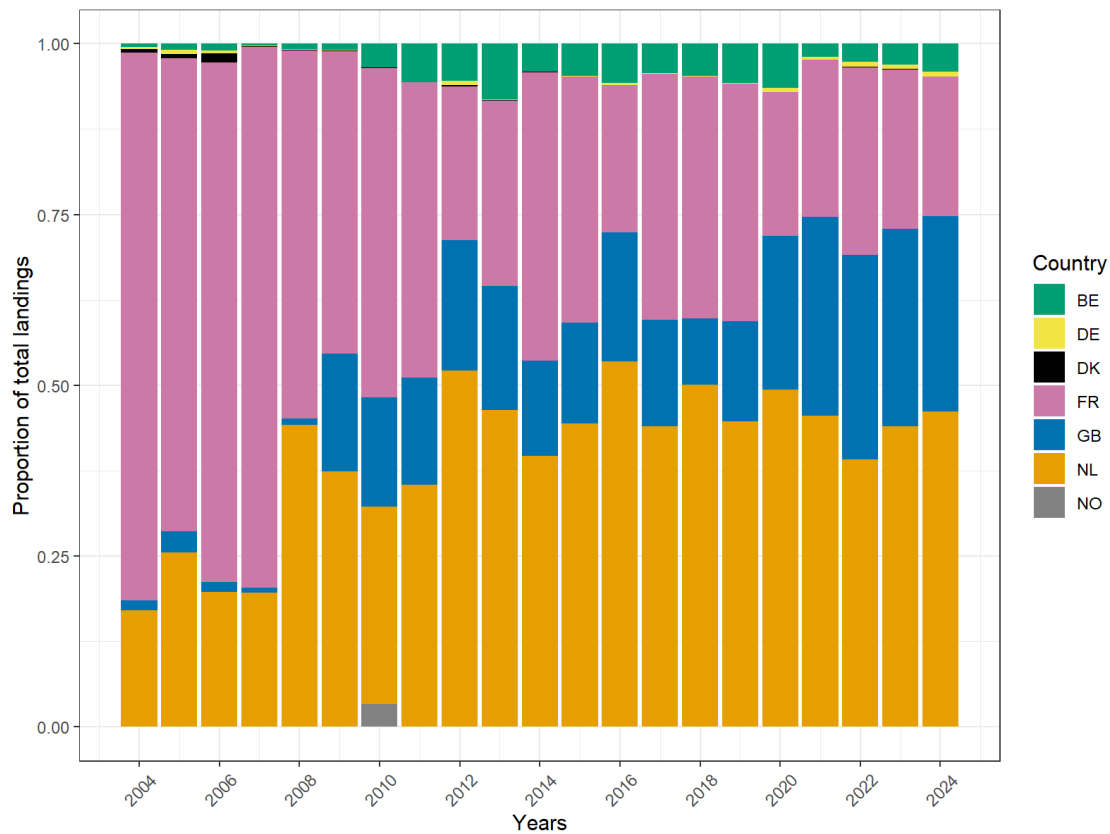


Figure 18.5.1.4. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Proportions of total ICES landings by country.

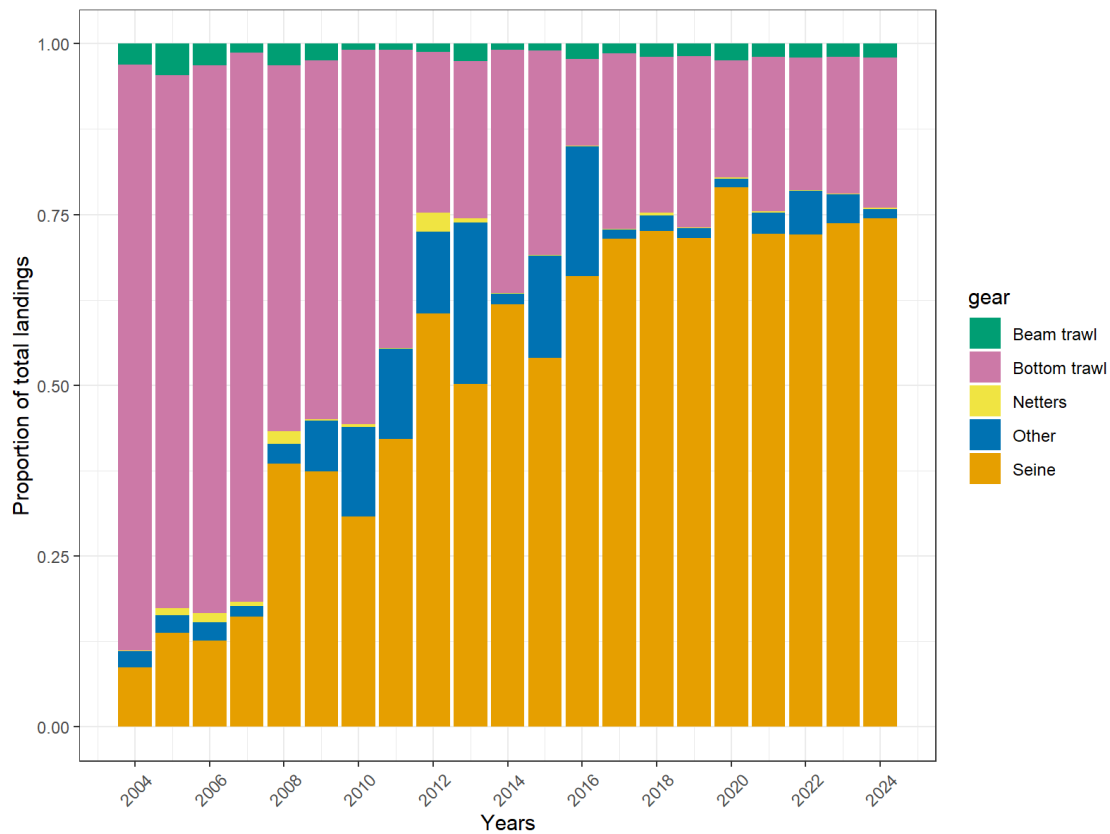


Figure 18.5.1.5. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Proportions of total ICES landings by gear.

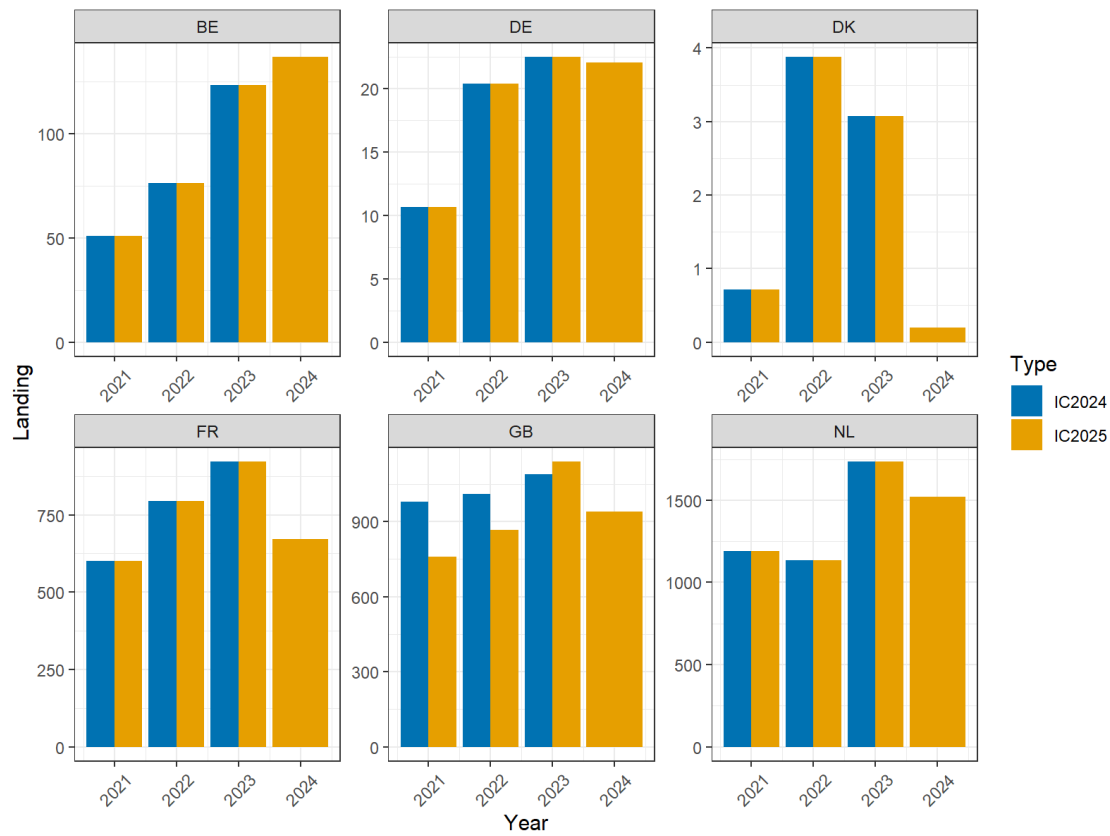


Figure 18.5.1.6. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. ICES landings submitted to InterCatch in 2024 versus 2025 per country.

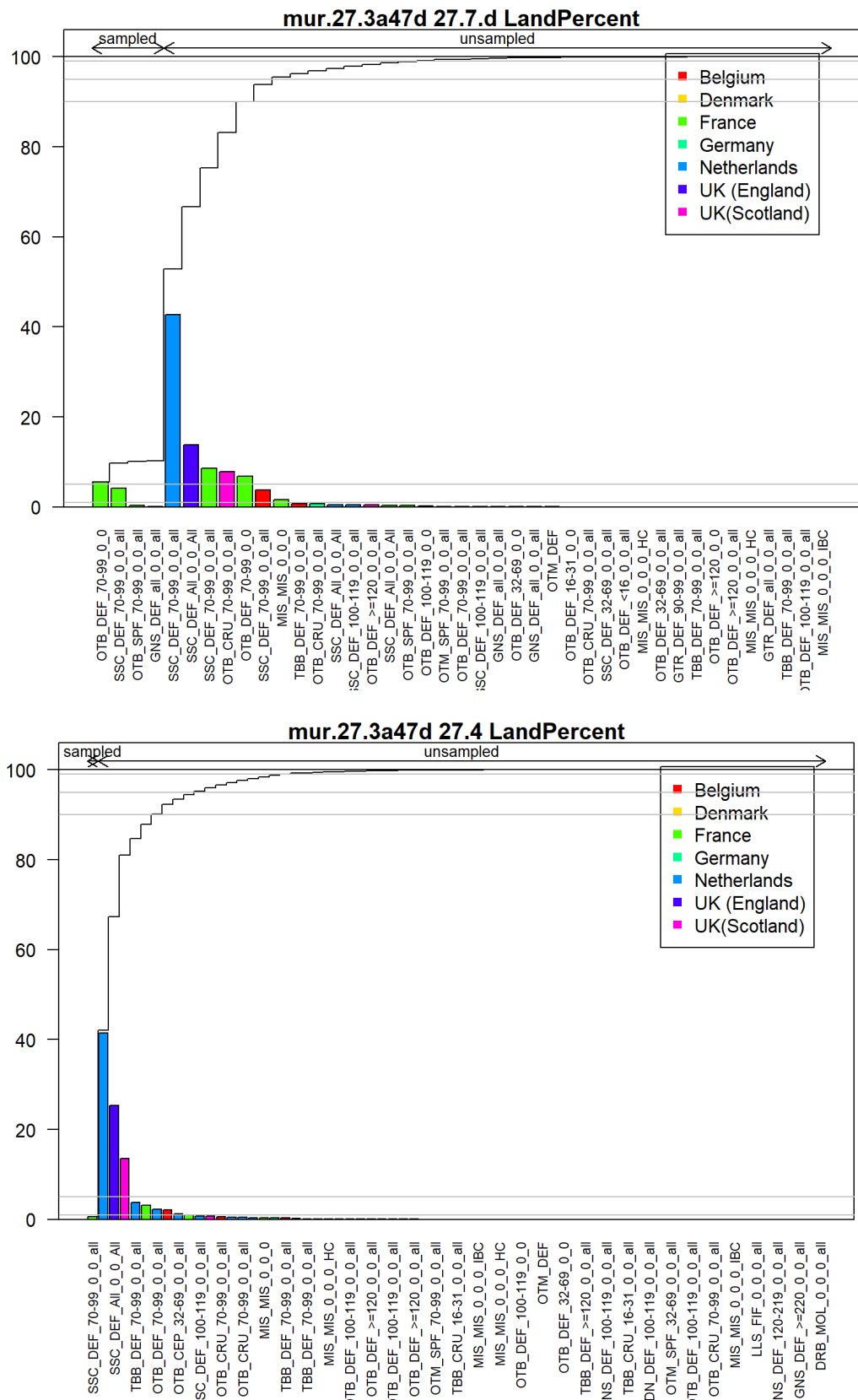


Figure 18.5.2.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. ICES landings with age samples by country and fleet for Division 7.d (upper panel) and Subarea 4 (lower panel) (percentage over the total area). No samples are provided for division 3.a.

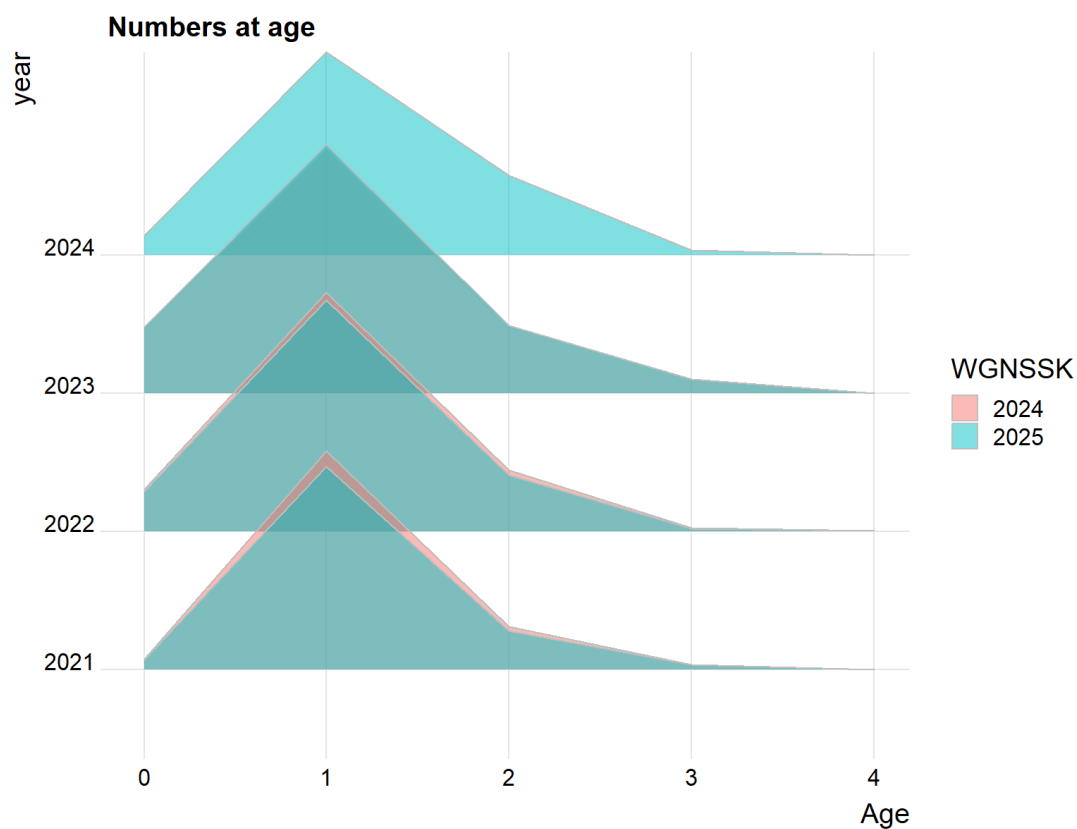


Figure 18.5.2.2. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Comparison of ICES landing number at age estimated during WGNSSK 2024 *versus* WGNSSK 2025 following data resubmission to InterCatch in 2025.

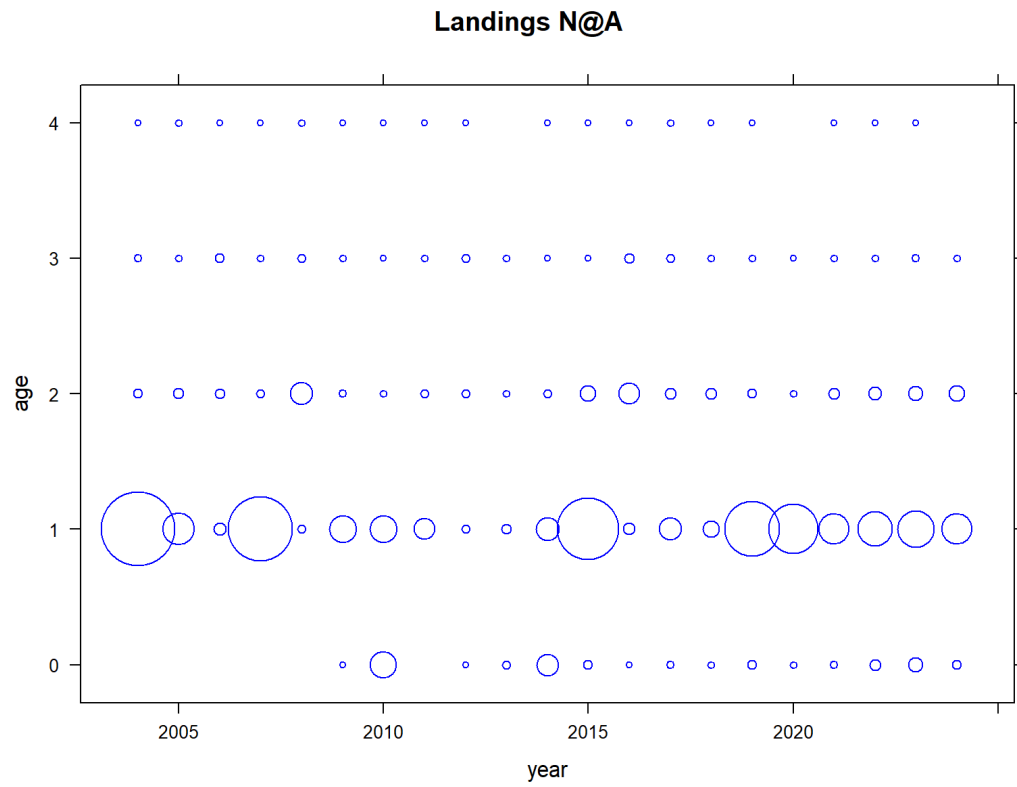


Figure 18.5.2.3. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Age structure (in numbers) as provided in the landings.

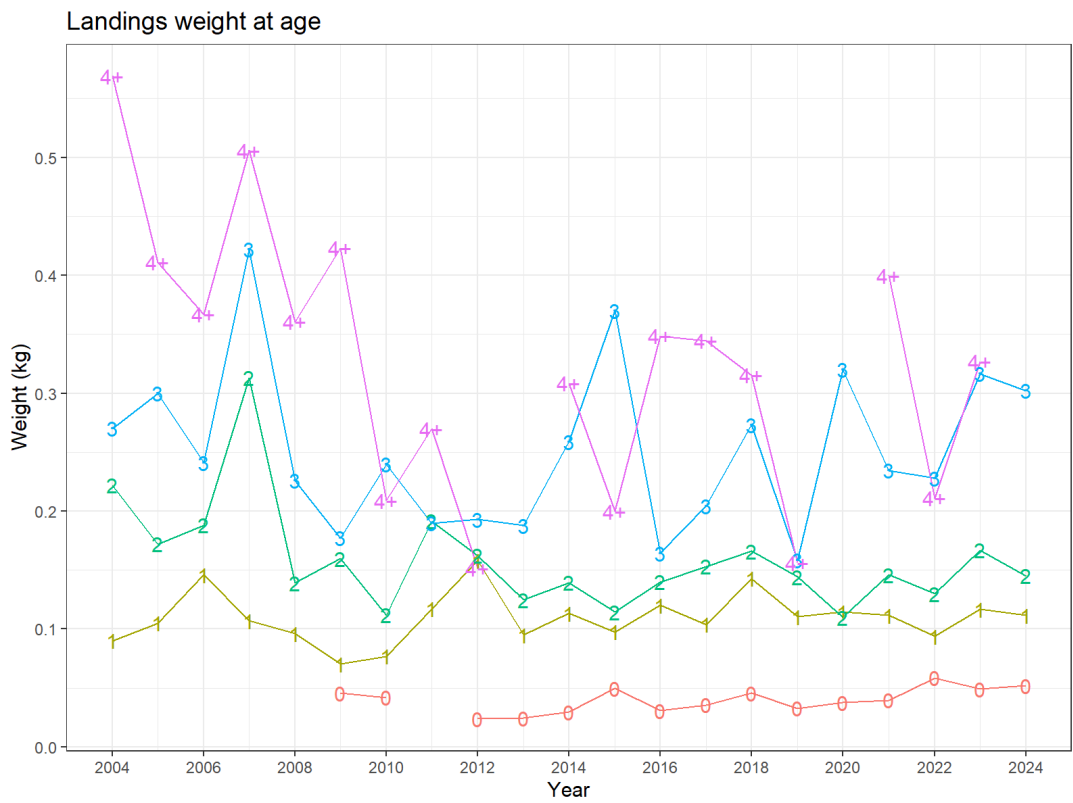


Figure 18.5.2.4. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Weight at age in the landings for age 0–4+.

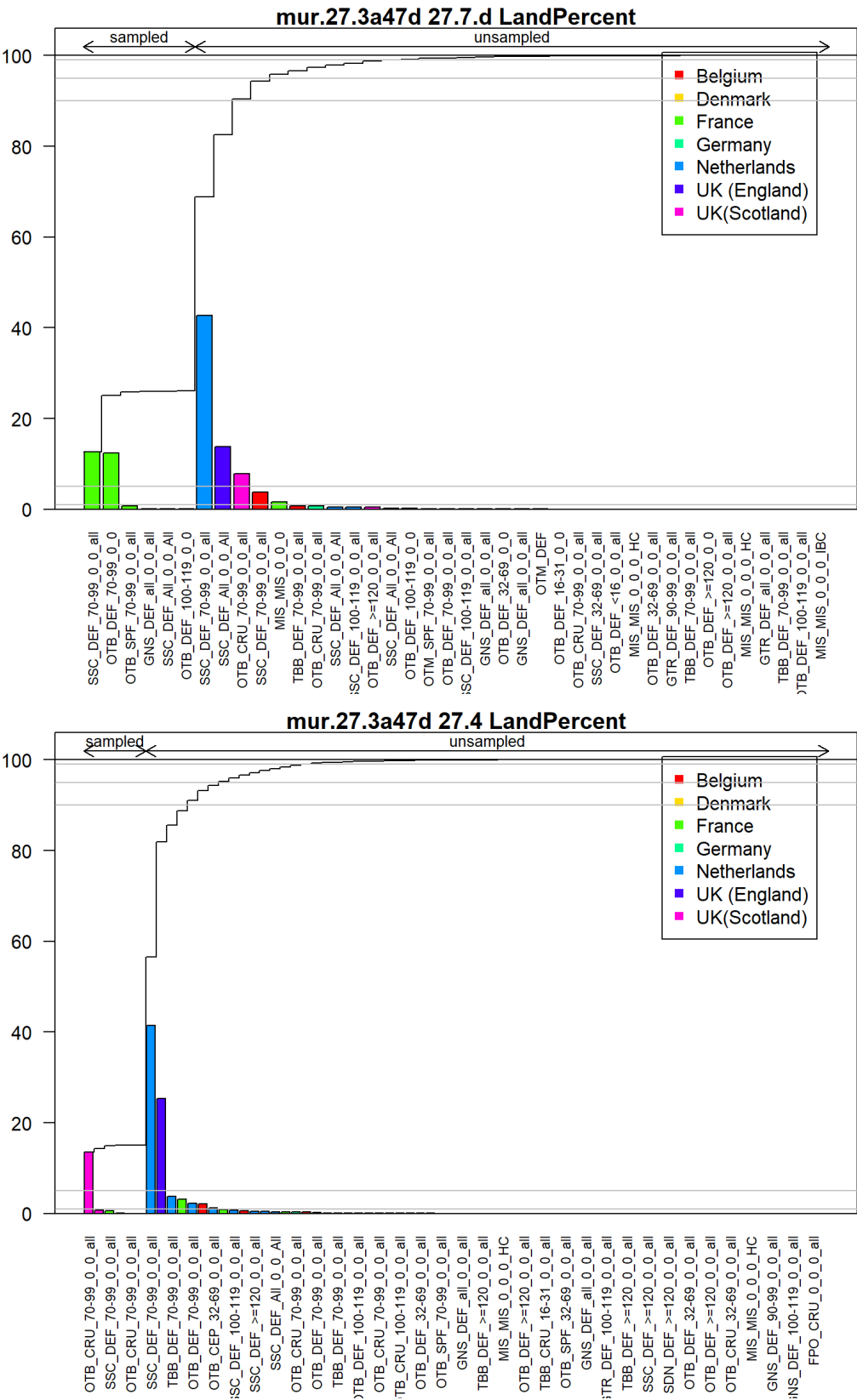


Figure 18.5.3.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. ICES landings with length samples by country and fleet for Division 7.d (upper panel) and Subarea 4 (lower panel; percentage over the total area). No samples are provided for division 3.a.

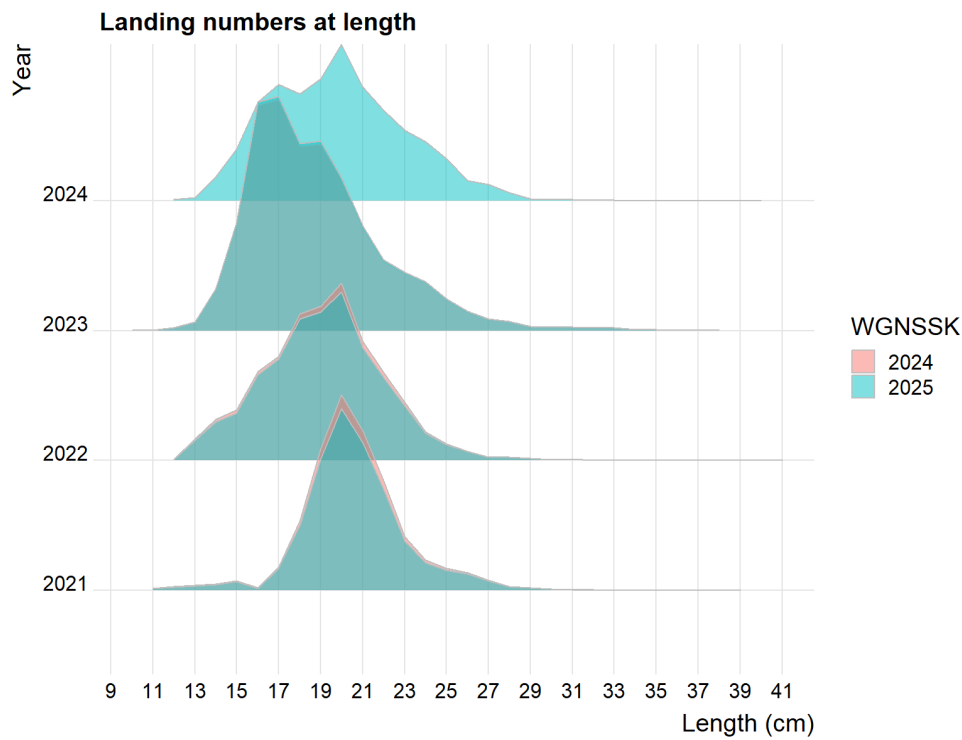


Figure 18.5.3.2. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Comparison of ICES landing numbers at length estimated during WGNSSK 2024 versus WGNSSK 2025 following data resubmission to InterCatch in 2025.

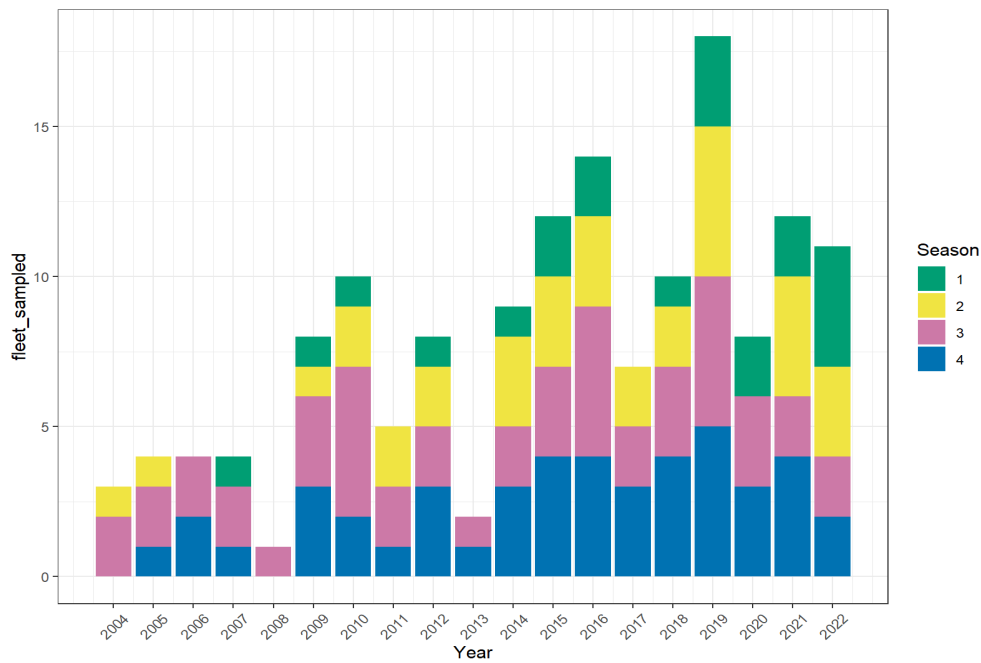


Figure 18.5.3.3. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Number of fleets with length samples in the French national database by year and quarter.

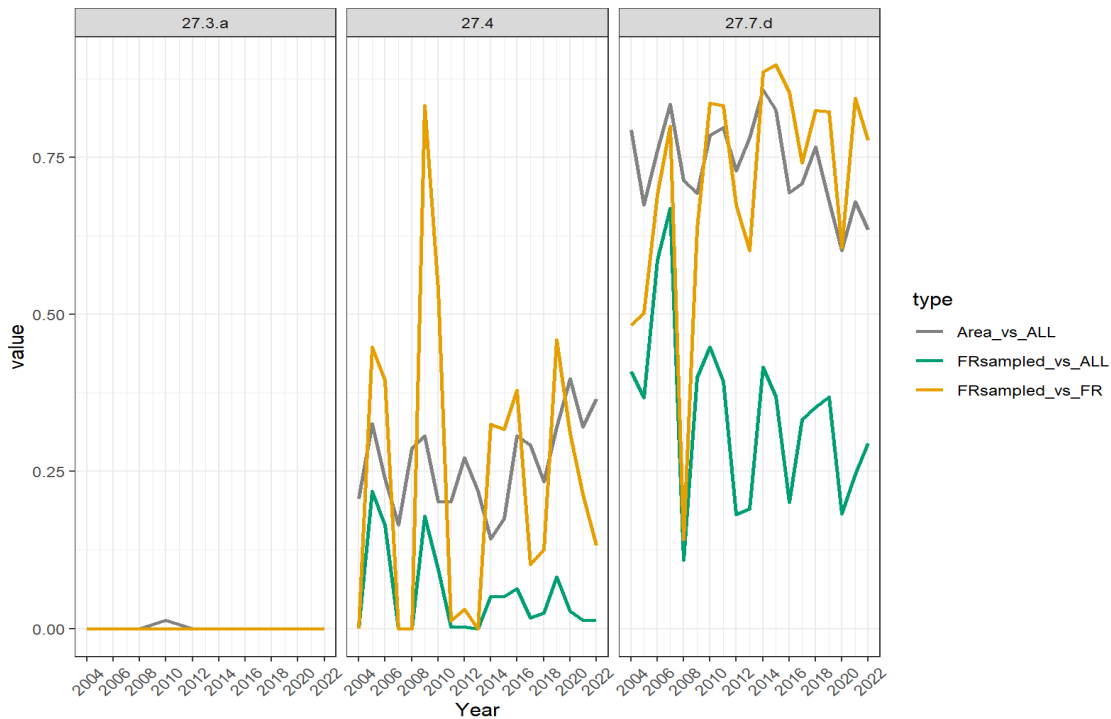


Figure 18.5.3.4. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Landings proportion by area and year. The orange line represents the proportion of French landings with samples compared to total French landings, the green line represents the proportion of French landings with samples compared to total ICES landings and the grey line shows the proportion of ICES landings in each area compared to ICES landings of the stock.

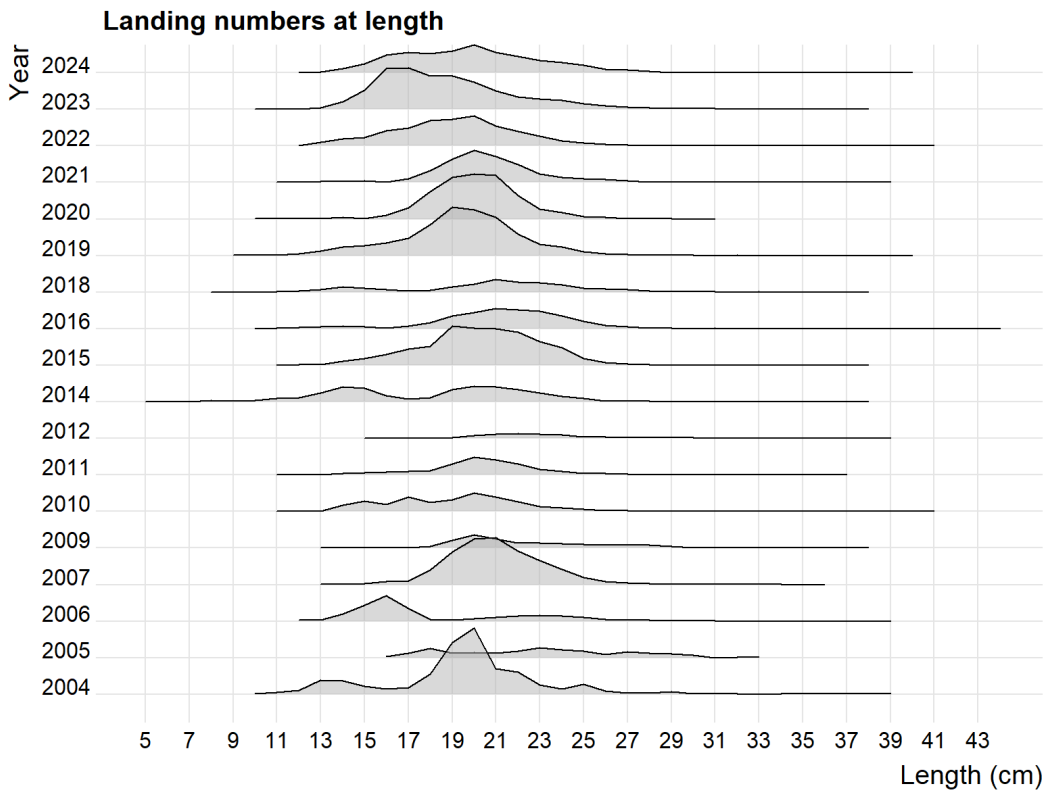


Figure 18.5.3.5. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Length distribution of landings estimated from the French national database prior to 2014 and from InterCatch thereafter.

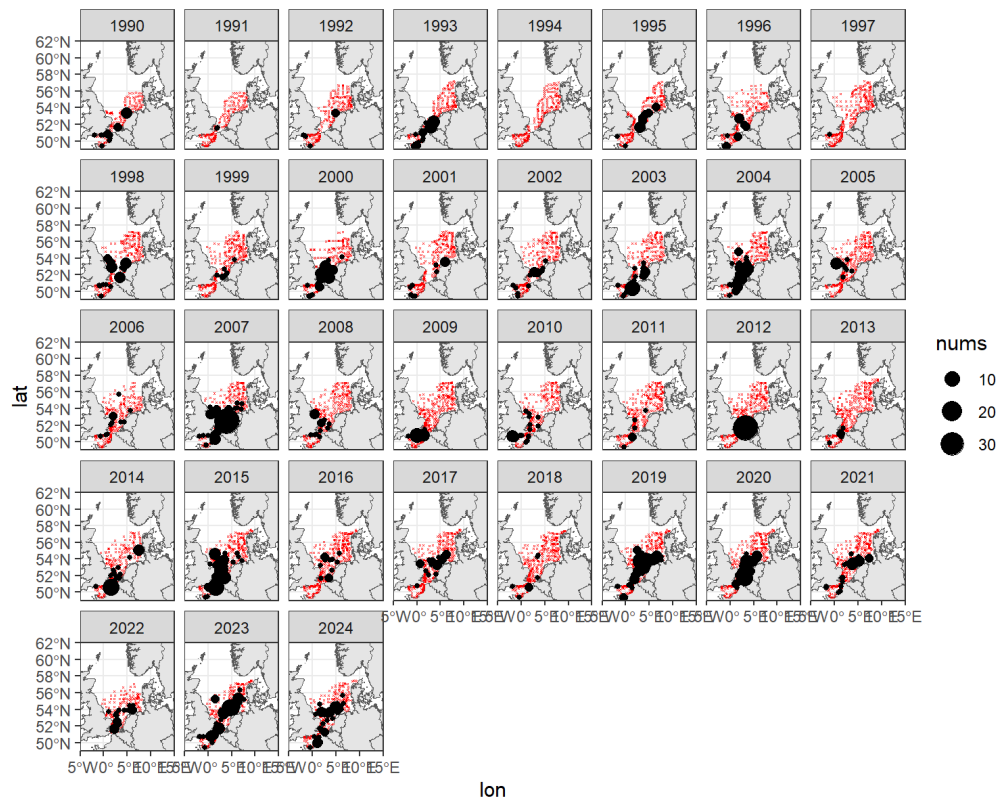


Figure 18.5.4.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. BTS Q3 abundance distribution of striped red mullet larger than 12 cm. Red crosses represent survey stations.



Figure 18.5.4.2. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. NS-IBTS Q3 abundance distribution of striped red mullet larger than 12 cm. Red crosses represent survey stations.

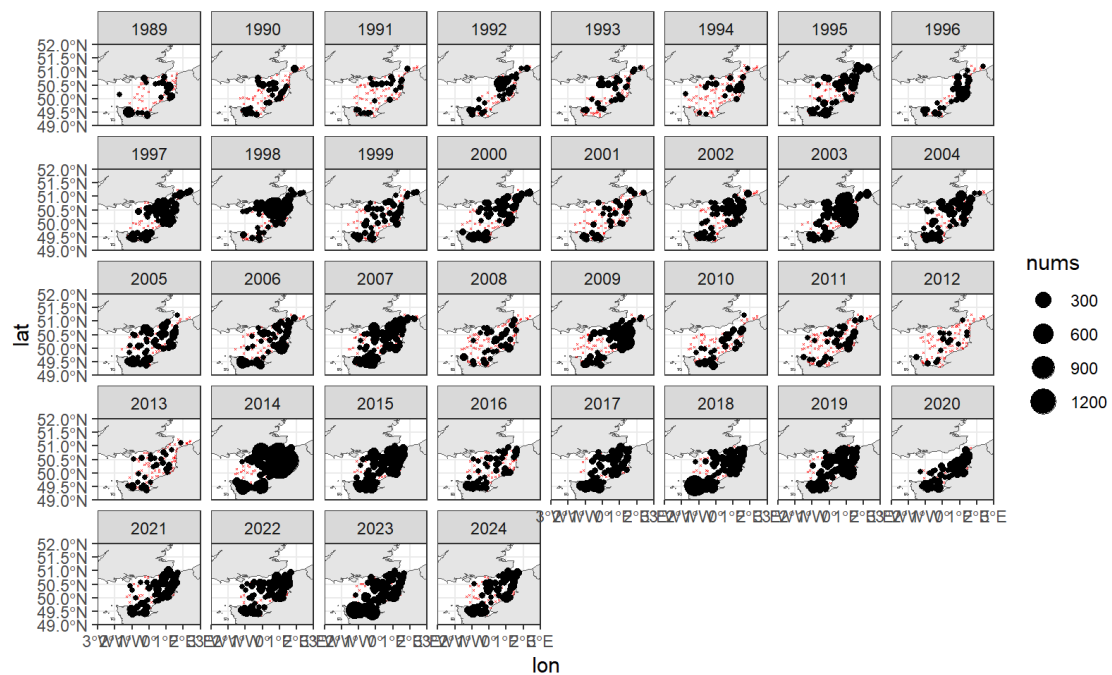


Figure 18.5.4.3. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. FR-GFS Q4 abundance distribution of striped red mullet larger than 12 cm. Red crosses represent survey stations.

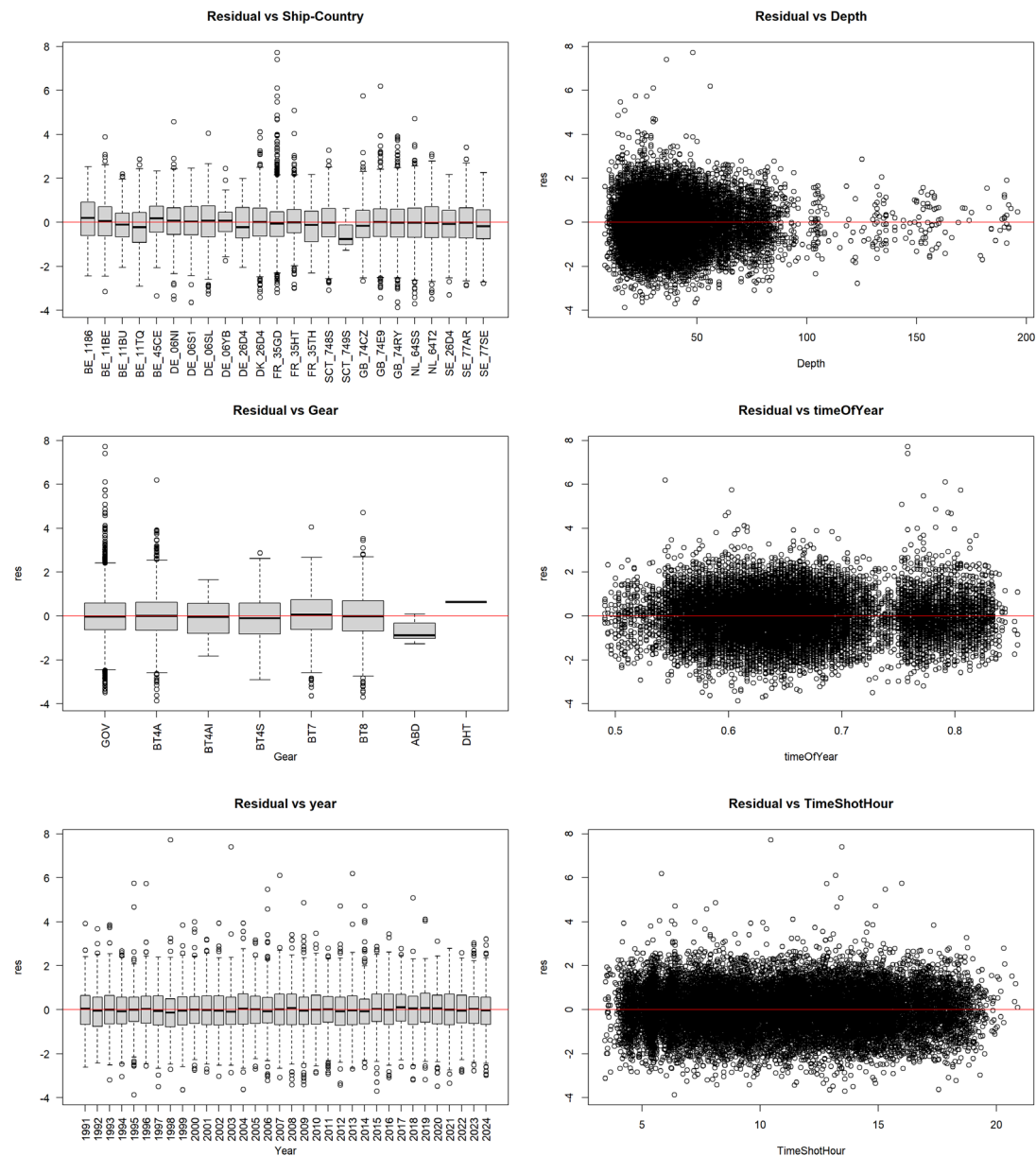


Figure 18.5.4.4 Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Survey index model residuals against covariates.

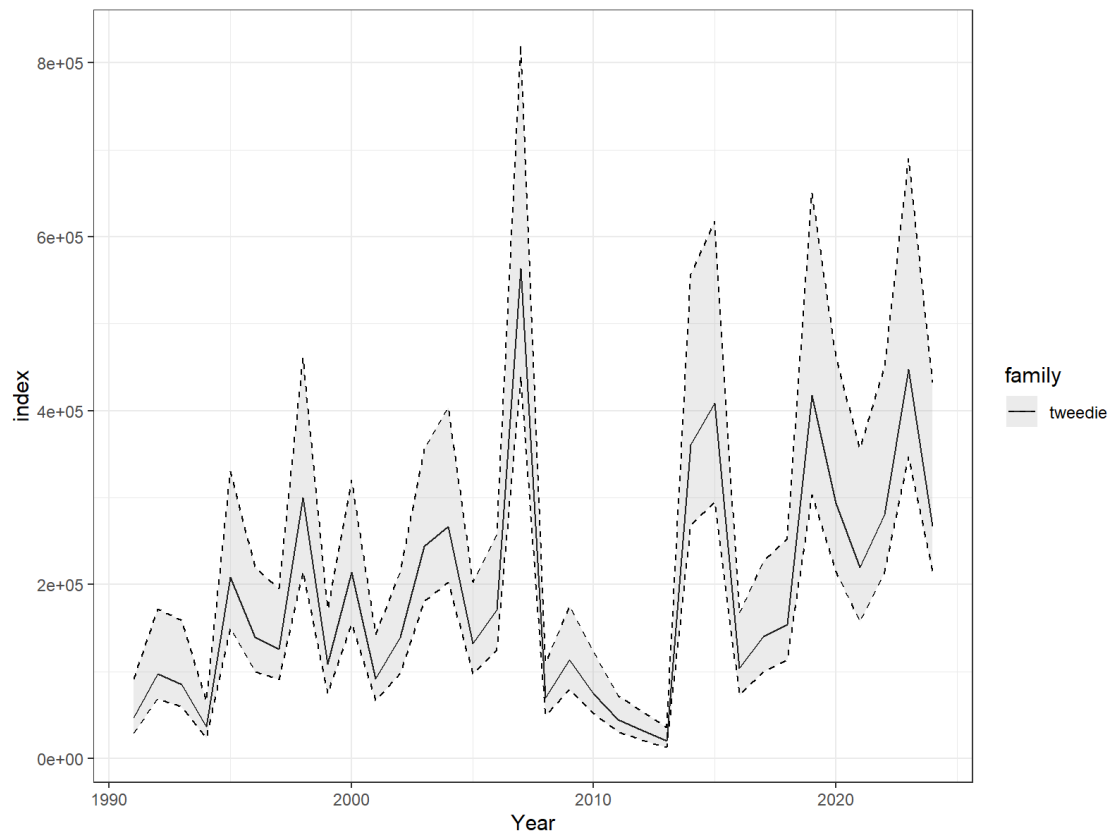


Figure 18.5.4.5 Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Q3 – Q4 Surveys exploitable biomass index using data up to 2024.

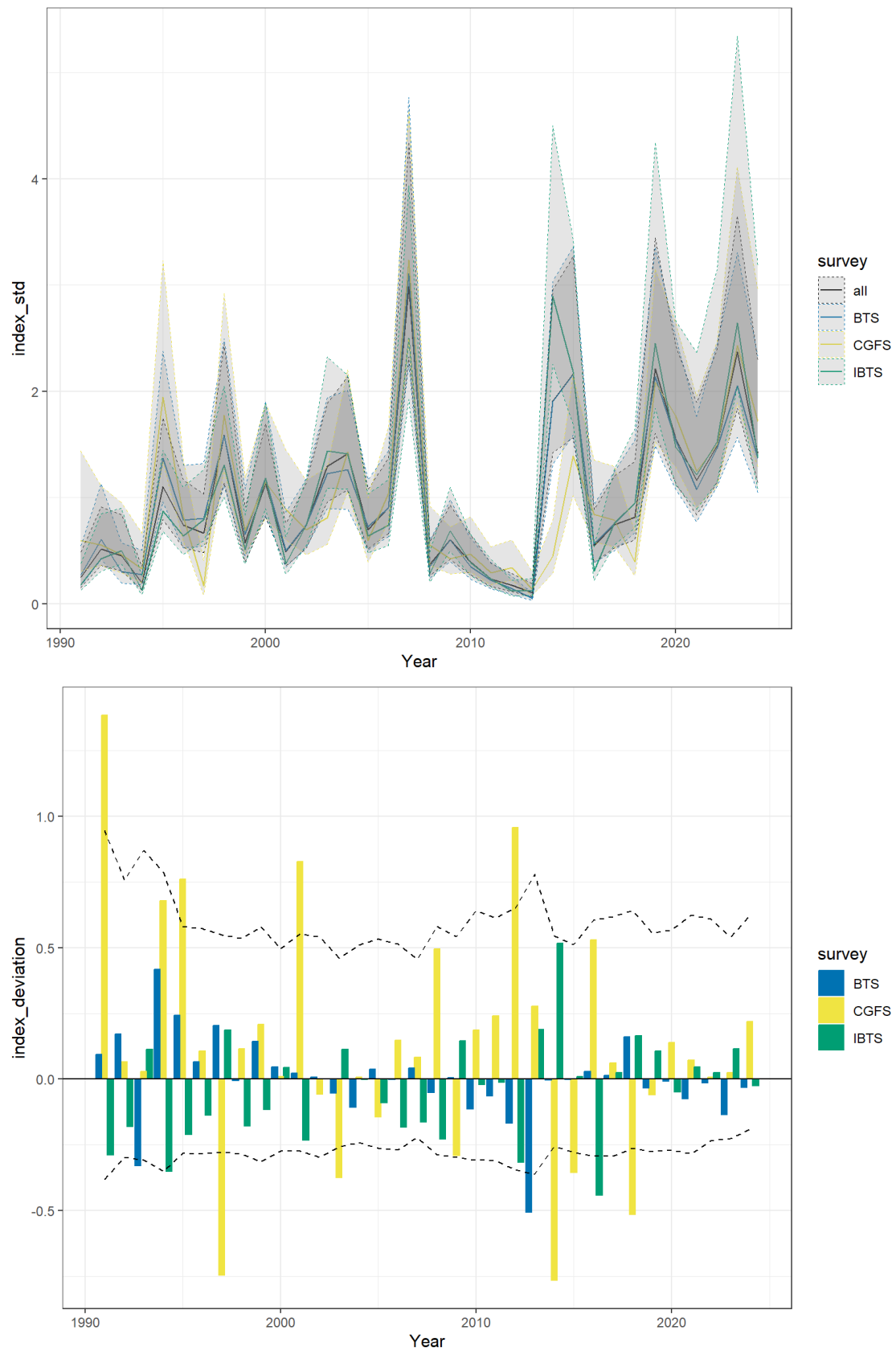


Figure 18.5.4.6 Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Q3 – Q4 Surveys exploitable biomass index leave-one-out. The top figure shows a comparison of trends and uncertainty, the bottom figure illustrates the proportion of deviation from the combined-index Q3-Q4 IBTS, BTS and CGFS (black line), the dashed lines show index uncertainties.

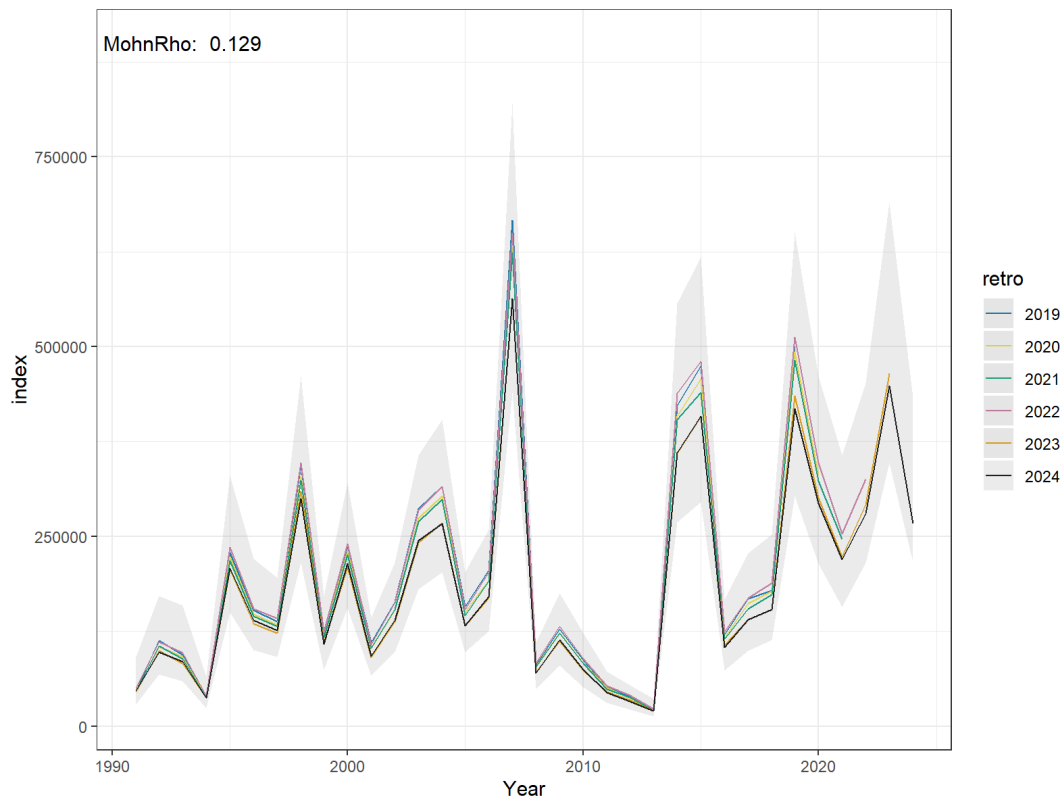


Figure 18.5.4.7 Striped red mullet in Subarea 4 and divisions 7.d and 3.a. Q3 – Q4 Surveys exploitable biomass index retrospective up to 2024.

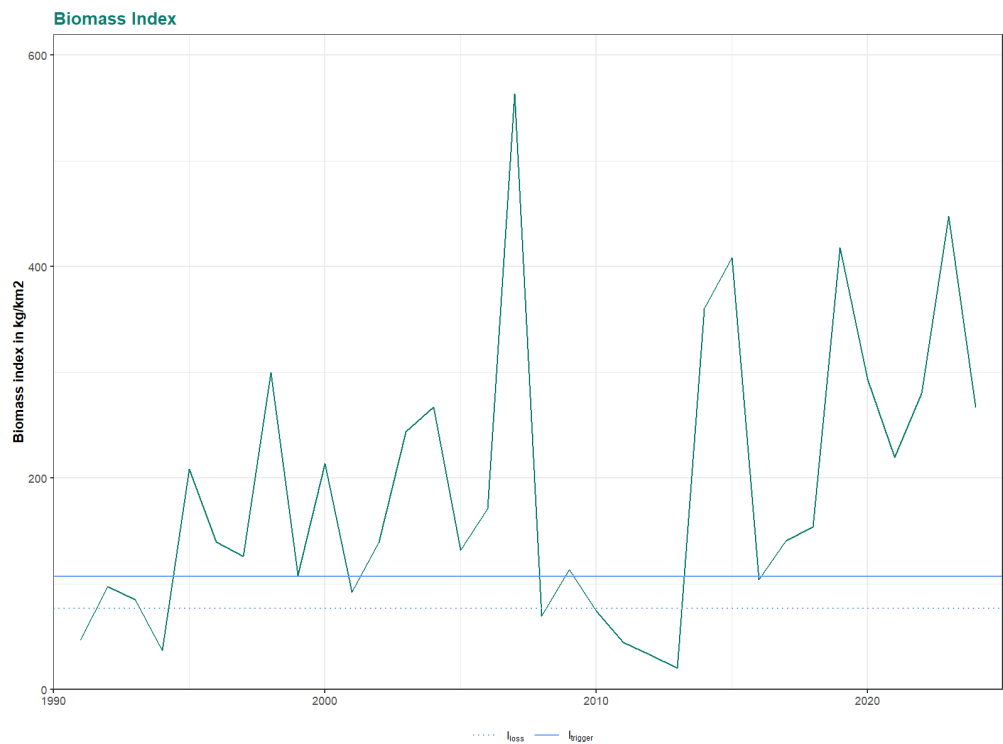


Figure 18.6.1.1. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. WGNSSK 2025 basis of ICES advice for 2026–2027. Exploitable biomass index compared to l_{trigger} . l_{trigger} was updated using the same year range as during the benchmark (1991-2022).

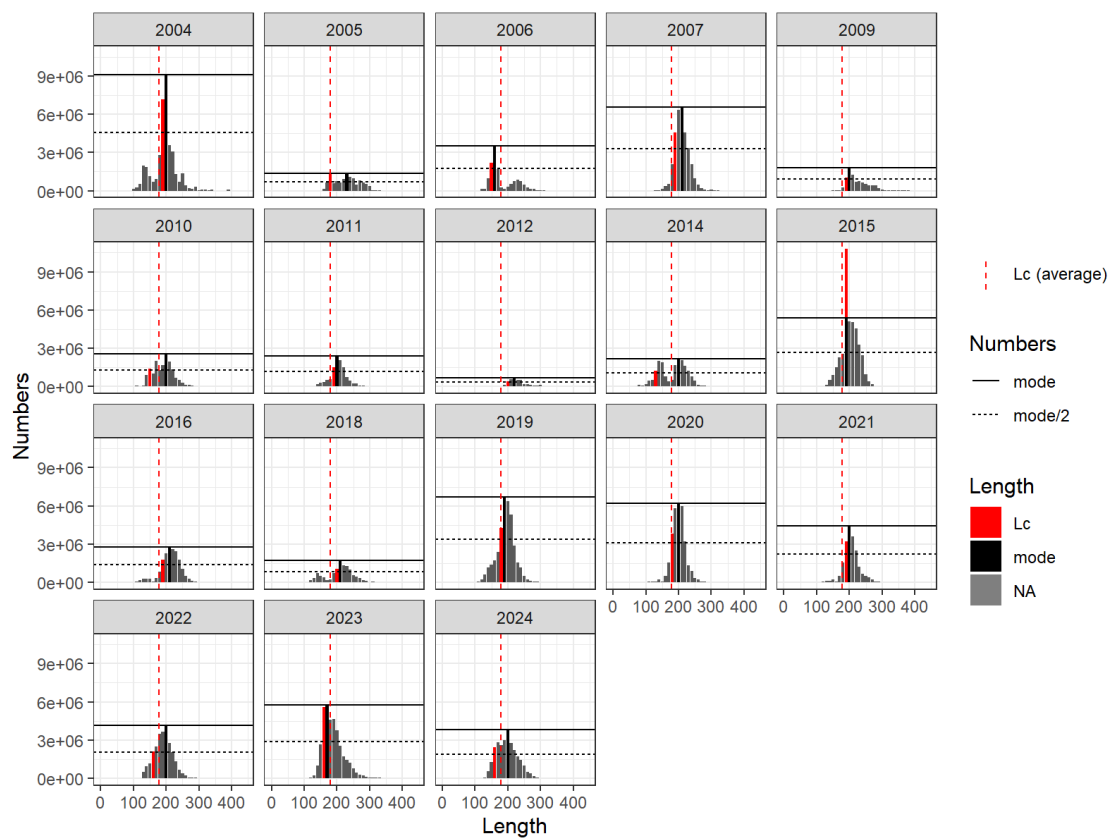


Figure 18.6.1.2. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. WGNSSK 2025 basis of ICES advice for 2026–2027. Commercial landings length frequencies, yearly length at first catch (Lc) and average Lc over 2004–2022.

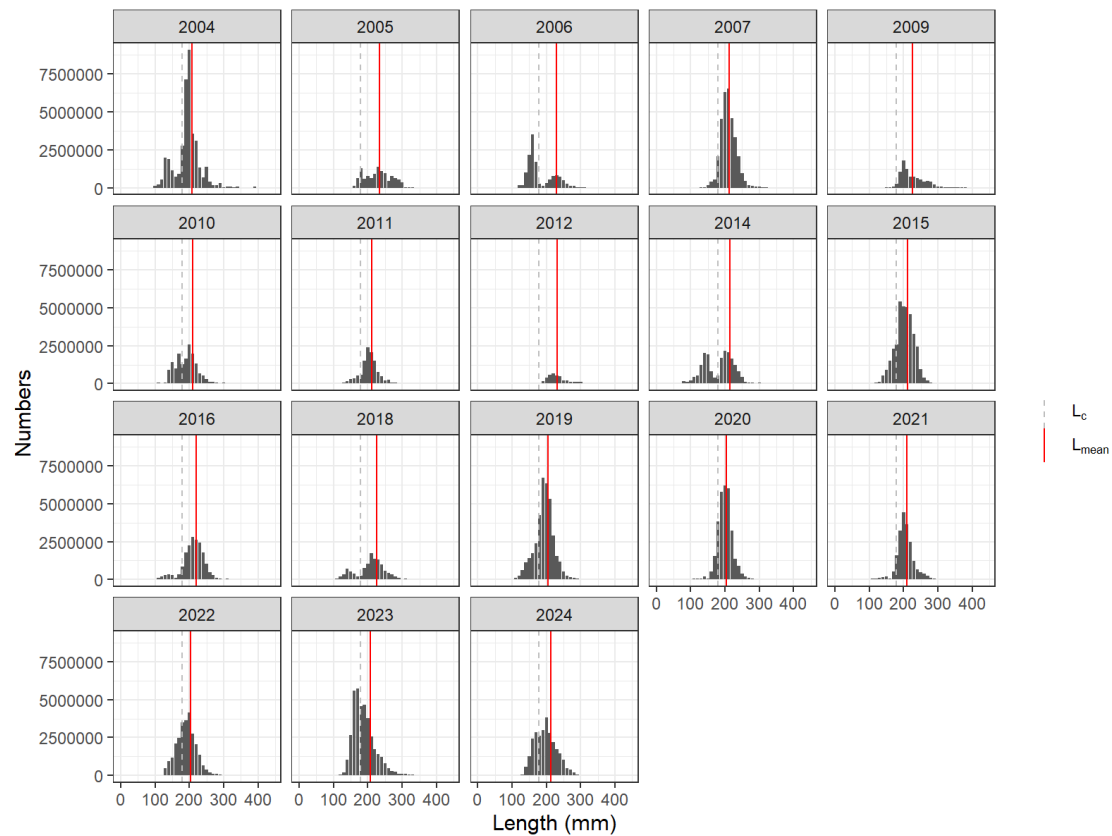


Figure 18.6.1.3. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. WGNSSK 2025 basis of ICES advice for 2026–2027. Commercial landings length frequencies, average length at first catch (L_c) and mean length above L_c (L_{mean}) over 2004–2022.

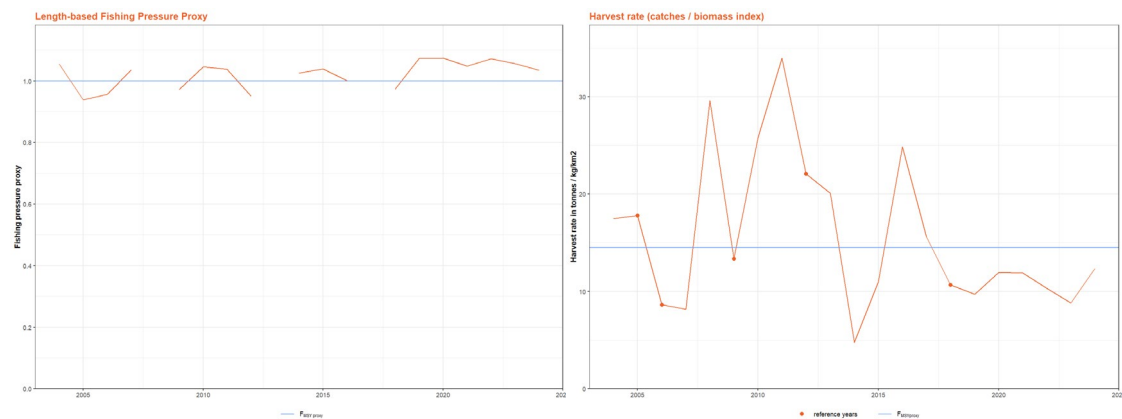


Figure 18.6.1.4. Striped red mullet in Subarea 4 and divisions 7.d and 3.a. WGNSSK 2025 basis of ICES advice for 2026–2027. $F_{MSY\text{ proxy}}$ indicator relative to the fishing pressure proxy ratio $L_F = M / L_{mean}$ (inverse f) and Harvest rate proxy (ratio of catches/ biomass index). Orange dots highlight years when length-based indicator inverse f is below 1 and the corresponding proxy harvest rates over the period 2004–2022. The blue line in the left panel presents the value of $F_{MSY\text{ proxy}}$ and on the right panel $HR_{MSY\text{ proxy}}$.

Contents

- 18 Striped red mullet the North Sea, eastern English Channel, Skagerrak, and Kattegat..... 660
 - 18.1 General..... 660
 - 18.2 Fisheries 661
 - 18.3 ICES historical advice 661
 - 18.4 Management..... 662
 - 18.5 Data available 662
 - 18.5.1 Catch 662
 - 18.5.2 Catch-at-age..... 663
 - 18.5.3 Catch-at-length 663
 - 18.5.4 Survey data 664
 - 18.6 Assessment 666
 - 18.6.1 Category 3 advice 2026–2027: *chr* rule (Method 2.2) 666
 - 18.6.2 2025 advice 670
 - 18.7 Length-based indicators screening 670
 - 18.8 Conclusions 671
 - 18.9 Issues list..... 671
 - 18.9.1 Data and stock ID 671
 - 18.9.2 Assessment 671
 - 18.9.3 Forecast and reference points 672
 - 18.10 References 672
 - 18.11 Figures and tables 673